Set 1A: AC Transfer Functions of RLC Networks

ECEN 325 — Lecture 1 (Linear Network Analysis) practice set

What this set trains. Finding $H(s) = V_o/V_i$ of a passive network with *voltage dividers only*, turning it into Bode-ready form, sketching the magnitude Bode plot, and computing the steady-state sinusoidal output. There are no DC sources here, so Step 2 of the pipeline is always trivial. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-1 pipeline (use it for every problem in this set)

Tools — this is all the theory you need:

T1 *Impedances:* $Z_R = R$, $Z_C = \frac{1}{sC}$, $Z_L = sL$. Series: add. Parallel: $Z_a \parallel Z_b = \frac{Z_a Z_b}{Z_a + Z_b}$.

T2 *Voltage divider:* if Z_{top} connects V_{in} to a node and Z_{bot} is everything from that node to ground, then $V_{node} = V_{in}\,\frac{Z_{bot}}{Z_{top}+Z_{bot}}$.

T3 *Divider between two voltages (DC):* a node tied to V_a through R_a and to V_b through R_b sits at $V = V_b + (V_a - V_b)\frac{R_b}{R_a+R_b}$, and looks like a source V with resistance $R_a \parallel R_b$ (its Thevenin pair).

T4 *Dependent sources:* a VCVS forces $V = kV_x$; a VCCS drives the current gV_x through the impedance Z hanging on its node, so $V = \pm gV_xZ$ (sign from the arrow).

T5 *Sinusoids:* $A\sin(\omega t + \theta) \;\rightarrow\; |H(j\omega)|\,A\sin\big(\omega t + \theta + \angle H(j\omega)\big)$.

T6 *Bode rules:* a pole at p bends the magnitude by $-20\,\text{dB/dec}$ above p, a zero by $+20\,\text{dB/dec}$; $\frac{1}{1+s/p}$ is $0\,\text{dB}$ below p; $\frac{s}{s+p}$ is $0\,\text{dB}$ above p.

Steps:

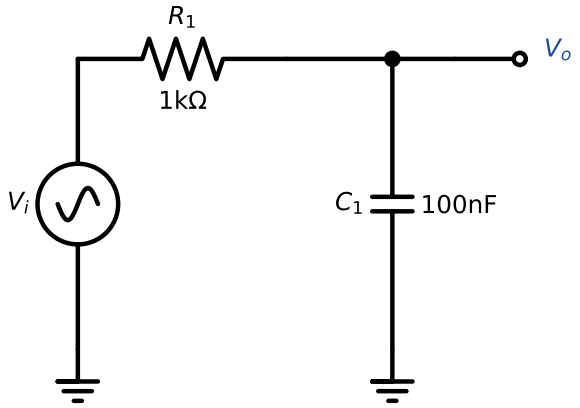
1. *Sources.* DC and AC both present $\Rightarrow$ superposition $V_o = V_{o,dc} + v_{o,ac}$.
2. *DC circuit:* $V_i \to 0$, $C \to$ open, $L \to$ short; reduce toward V_o with T3.
3. *AC circuit:* DC supplies $\to 0$ (rails become ground); merge parallel resistors.
4. *Divider chain:* name the nodes from V_i to V_o; $H = \prod$ (node-to-node ratios), each by T2 or T4, using the impedance seen looking right from each node.
5. *Standard form:* simplify, find the roots, write $H = K \cdot \prod(\text{factors})$.
6. *Bode:* start from the low-frequency slope, bend by $\pm 20\,\text{dB/dec}$ at each corner.
7. *Output:* evaluate each factor at $s = j\omega_{in}$, multiply magnitudes, add angles, apply T5 and add $V_{o,dc}$.

Problems

Easy problems

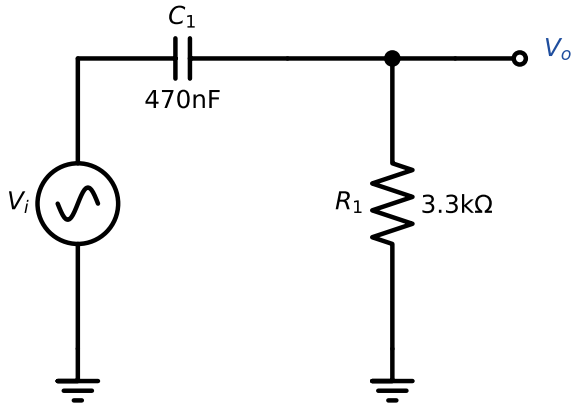
Problem 1A.1 **Easy**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



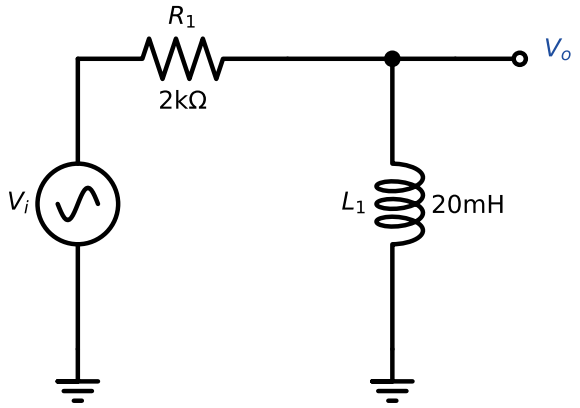
Problem 1A.2 **Easy**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 200\,t)$.



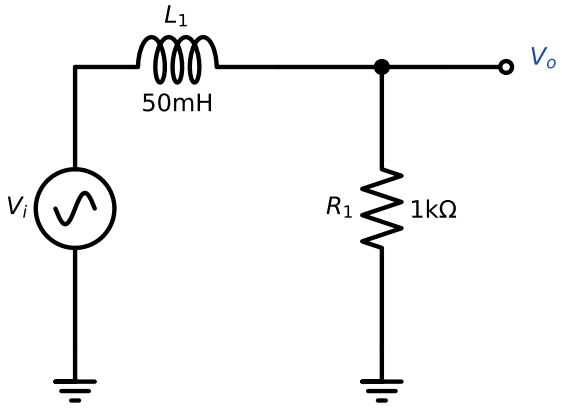
Problem 1A.3 **Easy**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 10000\,t)$.



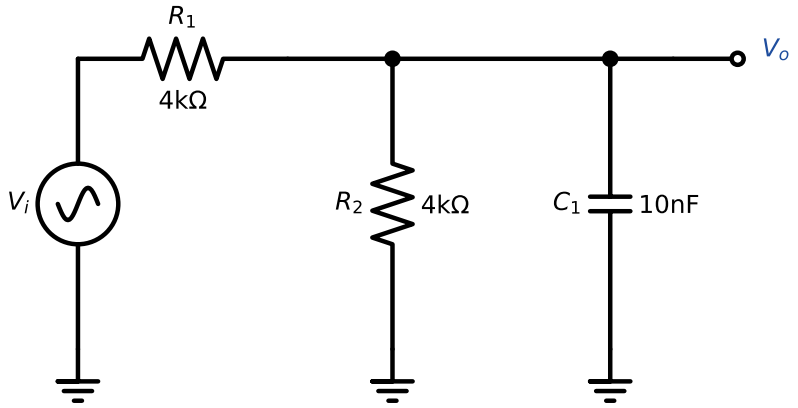
Problem 1A.4 **Easy**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 3\cos(2\pi \cdot 5000\,t)$.



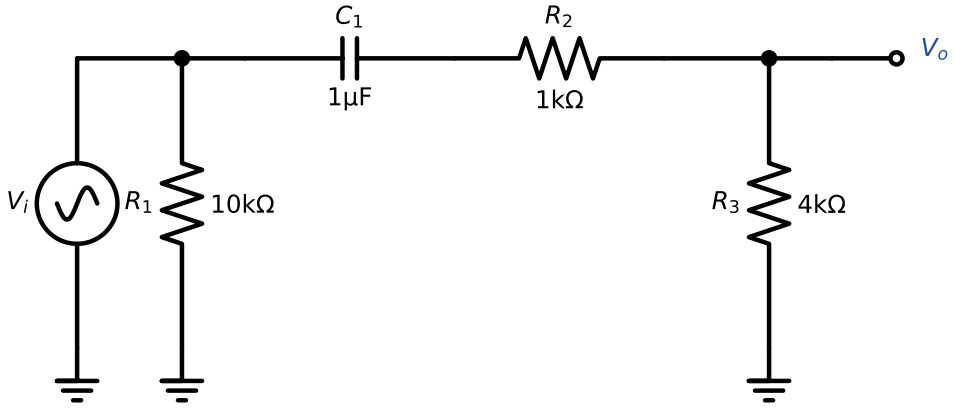
Problem 1A.5 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 2000\,t)$.



Problem 1A.6 — Easy

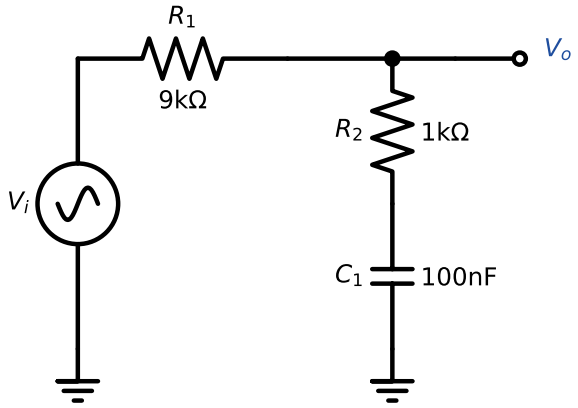
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 50\,t)$.



Medium problems

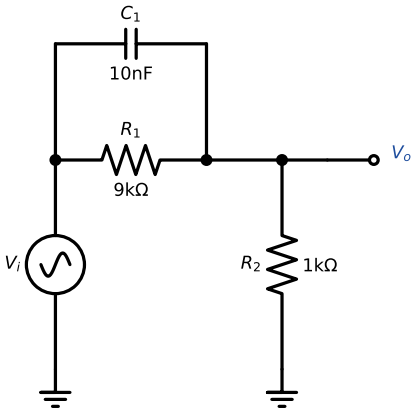
Problem 1A.7 — **Medium**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$.



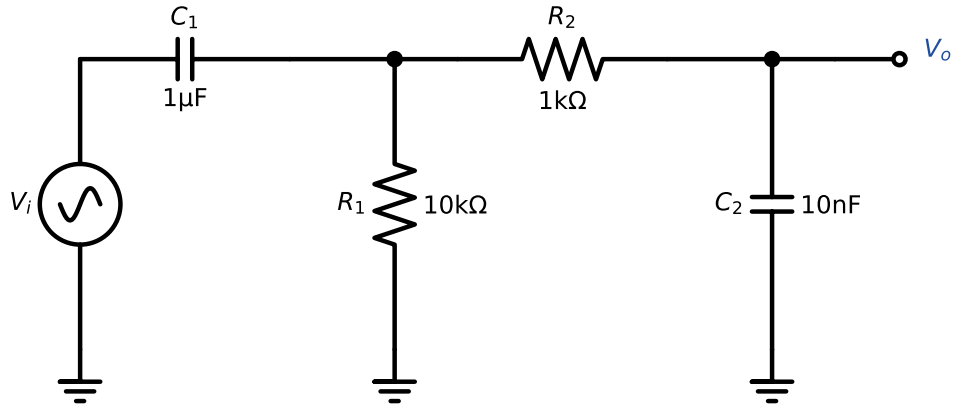
Problem 1A.8 — **Medium**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 5000\,t)$.



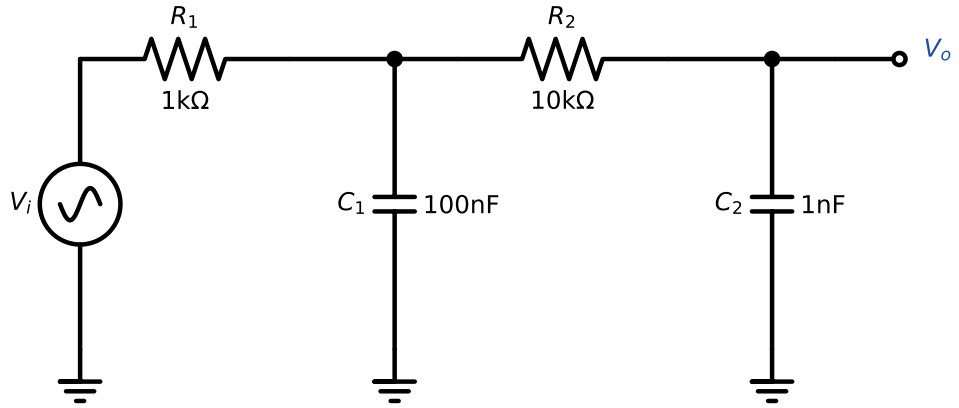
Problem 1A.9 **Medium**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



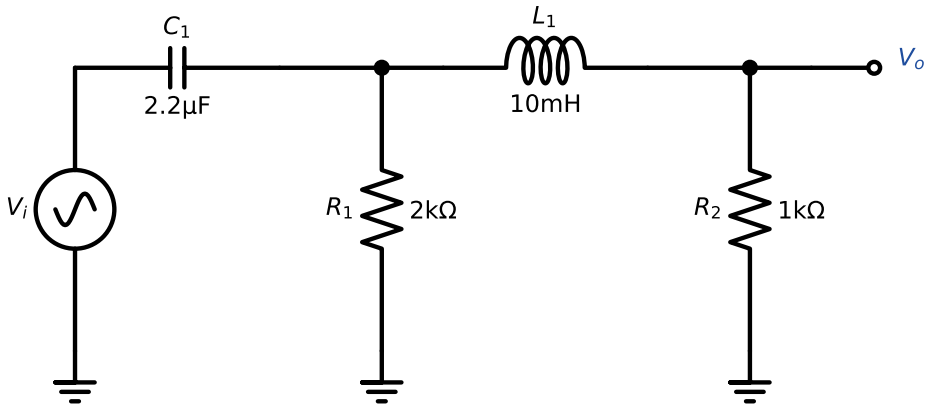
Problem 1A.10 **Medium**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 3000\,t)$.



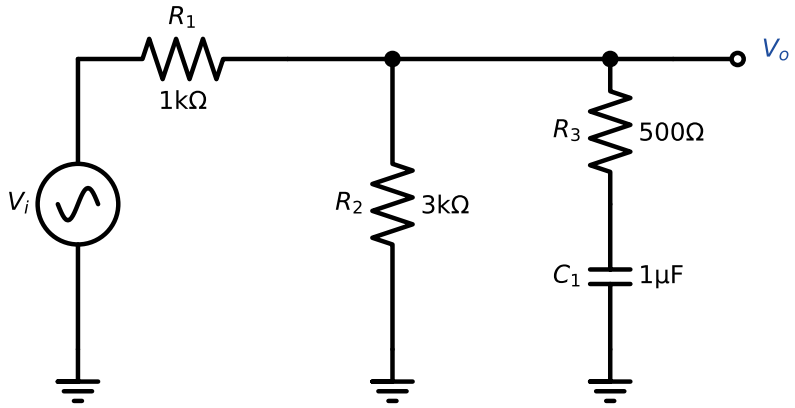
Problem 1A.11 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$.



Problem 1A.12 — Medium

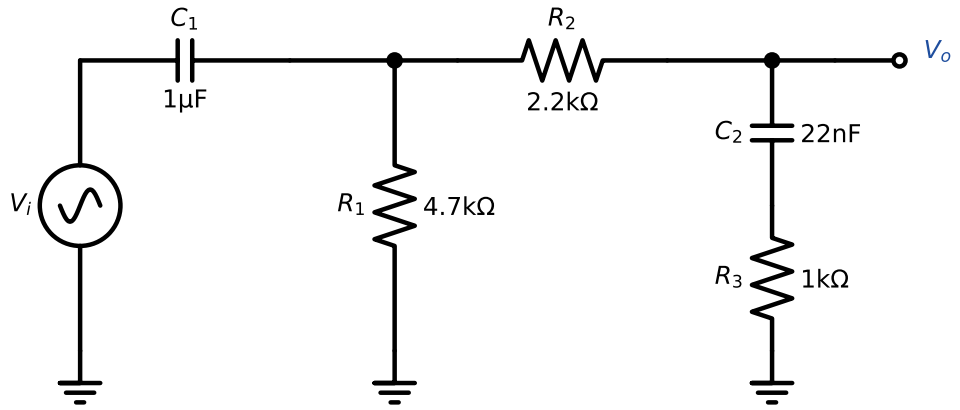
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



Hard problems

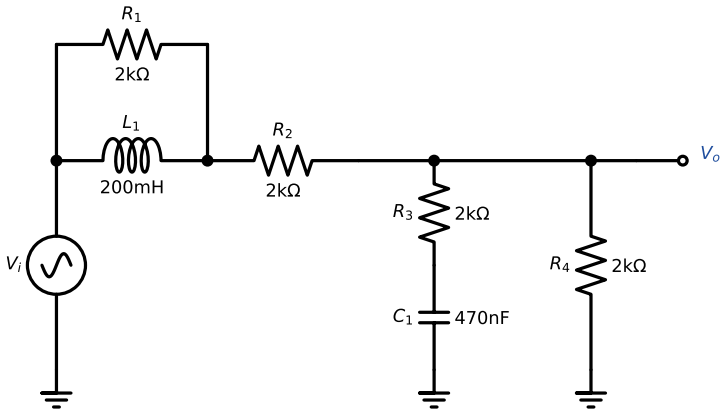
Problem 1A.13 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\cos\left(2\pi \cdot 2000\,t + \frac{\pi}{4}\right)$.



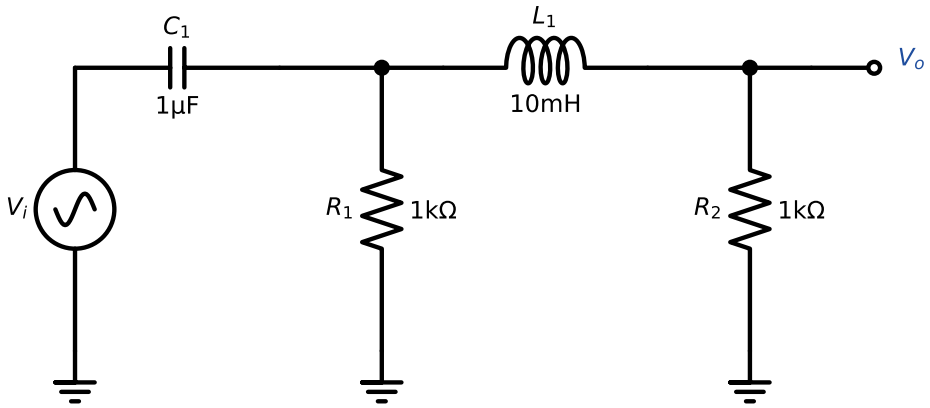
Problem 1A.14 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin\left(2\pi \cdot 1000\,t + \frac{\pi}{3}\right)$.



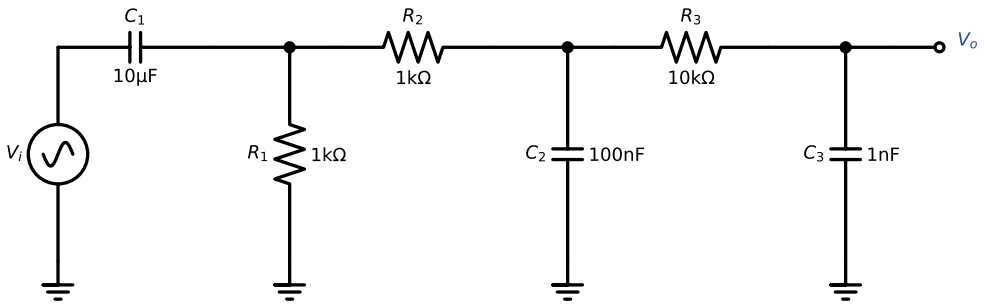
Problem 1A.15 **Hard**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 2000\,t)$.



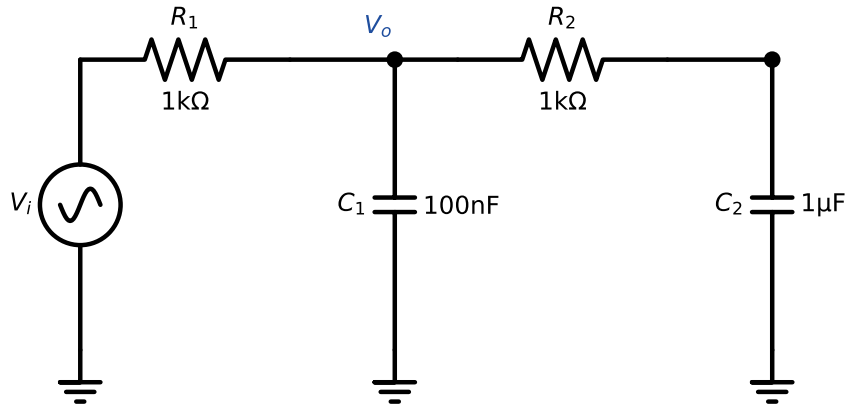
Problem 1A.16 **Hard**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$.



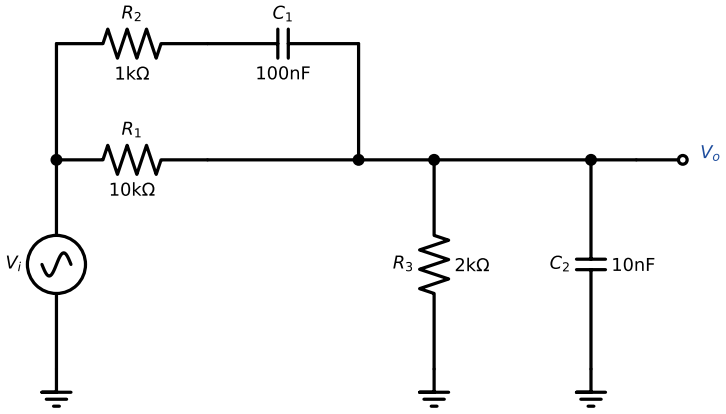
Problem 1A.17 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 500\,t)$. Note that V_o is *not* at the right end: the branch to its right loads it.



Problem 1A.18 — Hard

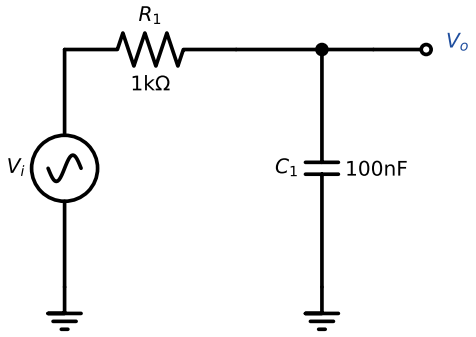
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin\left(2\pi \cdot 2000\,t - \frac{\pi}{6}\right)$.



Solutions

Every solution follows the same seven steps of the pipeline card.

Solution 1A.1 **Easy**



Step 1 — Identify the sources

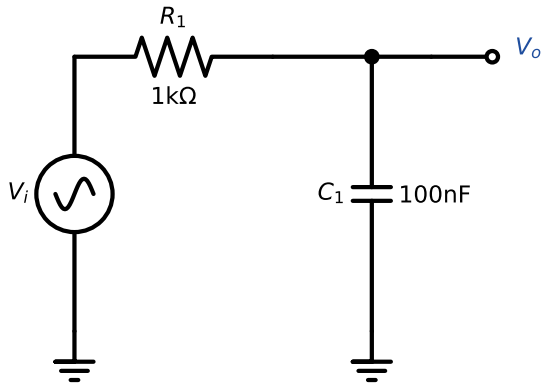
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{\frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{10000}{s + 10000}$$

So:

$$H(s) = \frac{10000}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = 10000\,\frac{1}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = 1$$

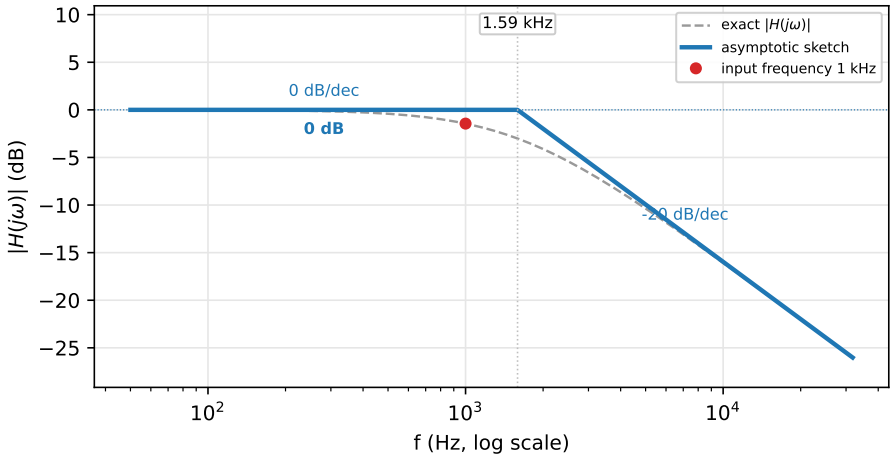
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	0 dB at end
$f > 1.59$ kHz	−20 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 2\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.8467	−0.561
total	**0.8467** (product)	**−0.561** (sum)

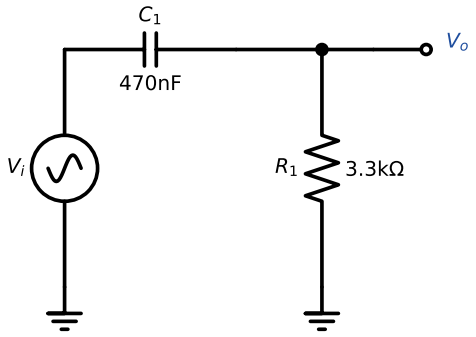
$$|H(j\omega_{in})| = 0.847, \qquad \angle H(j\omega_{in}) = -0.561 \text{ rad}$$
$$v_{o,ac}(t) = 0.847 \times 2\,\sin(2\pi \cdot 1000\,t - 0.561) = 1.69\,\sin(2\pi \cdot 1000\,t - 0.561)$$

$$\boxed{V_o(t) = 1.69\,\sin(2\pi \cdot 1000\,t - 0.561)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}} \quad \text{(corners: 1.59 kHz)}$$
$$V_o(t) = 1.69\,\sin(2\pi \cdot 1000\,t - 0.561)\ \text{V}$$

Solution 1A.2 **Easy**



Step 1 — Identify the sources

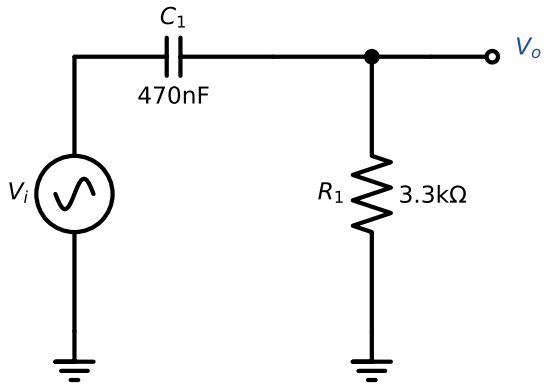
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_1}{\frac{1}{sC_1} + R_1} = \frac{s}{s + 645}$$

So:

$$H(s) = \frac{s}{s + 645}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 645$; $s + 645 = 0 \Rightarrow s = -645$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 645)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 103}, \qquad K = 1$$

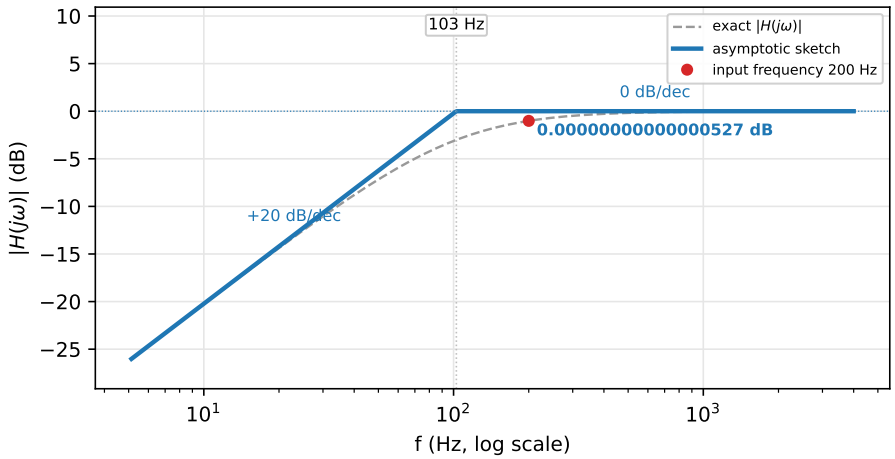
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	645	103 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00155\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 103$ Hz	+20 dB/dec	0 dB at end
$f > 103$ Hz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 200\,t)$, so $\omega_{in} = 2\pi \cdot 200 = 1260\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 103}$	0.8897	0.474
total	**0.8897** (product)	**0.474** (sum)

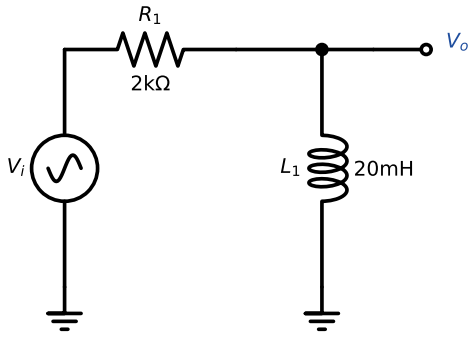
$$|H(j\omega_{in})| = 0.89, \qquad \angle H(j\omega_{in}) = 0.474 \text{ rad}$$
$$v_{o,ac}(t) = 0.89 \times 1 \sin(2\pi \cdot 200\,t + 0.474) = 0.89 \sin(2\pi \cdot 200\,t + 0.474)$$

$$\boxed{V_o(t) = 0.89 \sin(2\pi \cdot 200\,t + 0.474)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 103} \quad \text{(corners: 103 Hz)}$$
$$V_o(t) = 0.89 \sin(2\pi \cdot 200\,t + 0.474)\ \text{V}$$

Solution 1A.3 **Easy**



Step 1 — Identify the sources

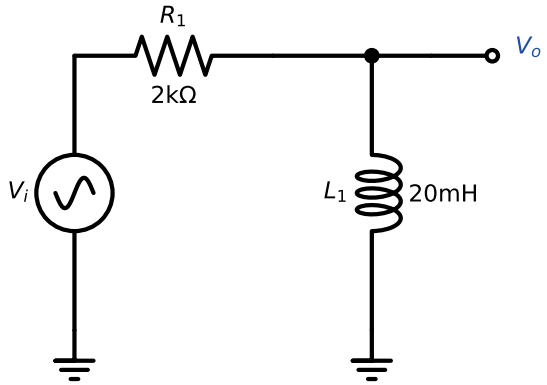
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{sL_1}{R_1 + sL_1} = \frac{s}{s + 10^5}$$

So:

$$H(s) = \frac{s}{s + 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 10^5$; $s + 10^5 = 0 \Rightarrow s = -10^5$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{s}{(s + 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 15.9\text{k}}, \qquad K = 1$$

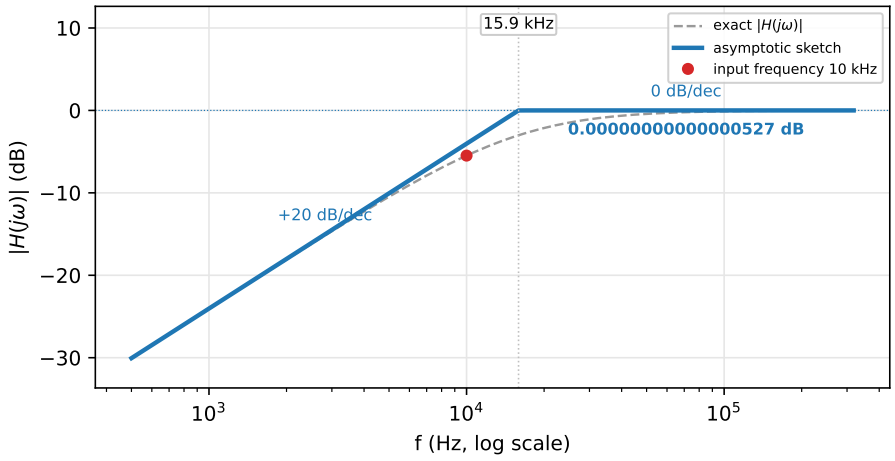
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10^5	15.9 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 10^{-5}\,s$, so the plot starts with a slope of $+20\,\text{dB/dec}$. Then, moving right, each pole bends the line down by $20\,\text{dB/dec}$ and each zero bends it up by $20\,\text{dB/dec}$ (tool T6):

frequency range	slope	asymptote level
$f < 15.9$ kHz	+20 dB/dec	0 dB at end
$f > 15.9$ kHz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 10000\,t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s + 2\pi \cdot 15.9\text{k}}$	0.532	1.01
total	**0.532** (product)	**1.01** (sum)

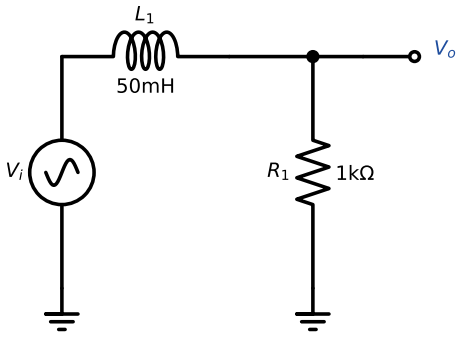
$$|H(j\omega_{in})| = 0.532, \qquad \angle H(j\omega_{in}) = 1.01 \text{ rad}$$
$$v_{o,ac}(t) = 0.532 \times 0.5 \sin(2\pi \cdot 10000\,t + 1.01) = 0.266 \sin(2\pi \cdot 10000\,t + 1.01)$$

$$\boxed{V_o(t) = 0.266 \sin(2\pi \cdot 10000\,t + 1.01) \text{ V}}$$

Answer

$$H(s) = 1\,\frac{s}{s + 2\pi \cdot 15.9\text{k}} \quad \text{(corners: 15.9 kHz)}$$
$$V_o(t) = 0.266 \sin(2\pi \cdot 10000\,t + 1.01) \text{ V}$$

Solution 1A.4 **Easy**



Step 1 — Identify the sources

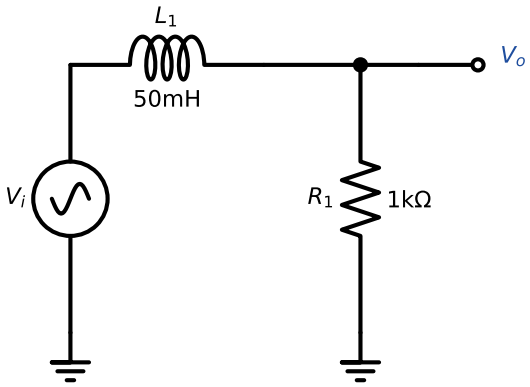
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_1}{sL_1 + R_1} = \frac{20000}{s + 20000}$$

So:

$$H(s) = \frac{20000}{s + 20000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 20000$; $s + 20000 = 0 \Rightarrow s = -20000$.

Factored form (leading coefficient in front):

$$H(s) = 20000\,\frac{1}{(s + 20000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}}, \qquad K = 1$$

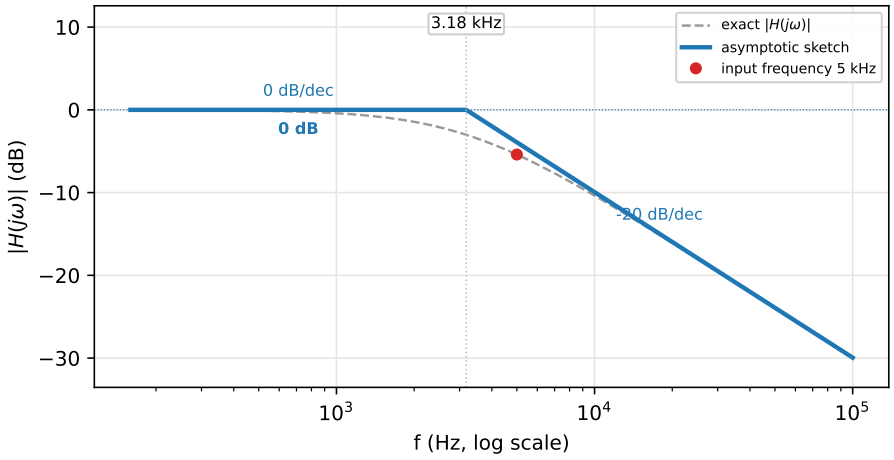
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	20000	3.18 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.18$ kHz	0 dB/dec (flat)	0 dB at end
$f > 3.18$ kHz	−20 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 3\cos(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}}$	0.537	-1
total	**0.537** (product)	**−1** (sum)

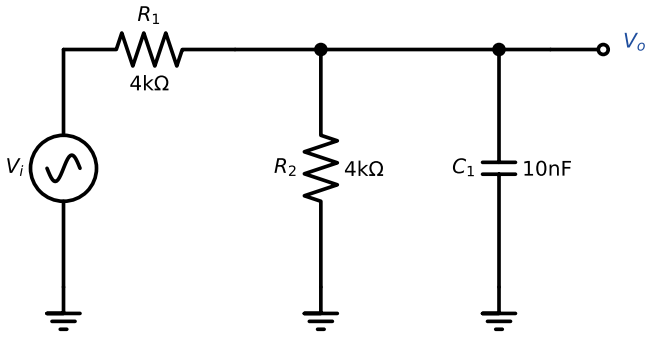
$$|H(j\omega_{in})| = 0.537, \qquad \angle H(j\omega_{in}) = -1 \text{ rad}$$
$$v_{o,ac}(t) = 0.537 \times 3\,\cos(2\pi \cdot 5000\,t - 1) = 1.61\,\cos(2\pi \cdot 5000\,t - 1)$$

$$\boxed{V_o(t) = 1.61\,\cos(2\pi \cdot 5000\,t - 1)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}} \quad \text{(corners: 3.18 kHz)}$$
$$V_o(t) = 1.61\,\cos(2\pi \cdot 5000\,t - 1)\ \text{V}$$

Solution 1A.5 **Easy**



Step 1 — Identify the sources

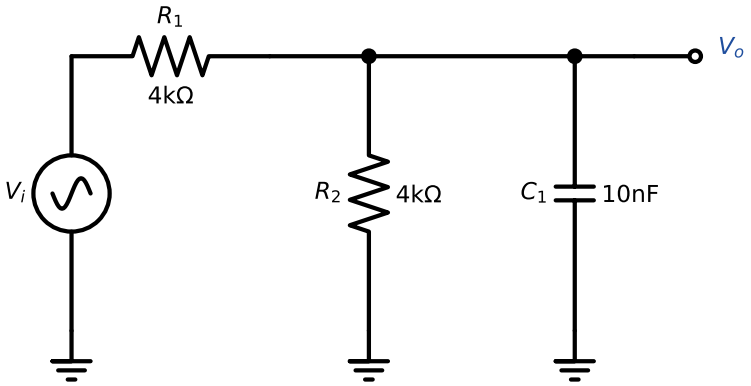
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 25000}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{25000}{s + 50000}$$

So:

$$H(s) = \frac{25000}{s + 50000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 50000$; $s + 50000 = 0 \Rightarrow s = -50000$.

Factored form (leading coefficient in front):

$$H(s) = 25000\,\frac{1}{(s + 50000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.5\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 7.96\text{k}}}, \qquad K = 0.5$$

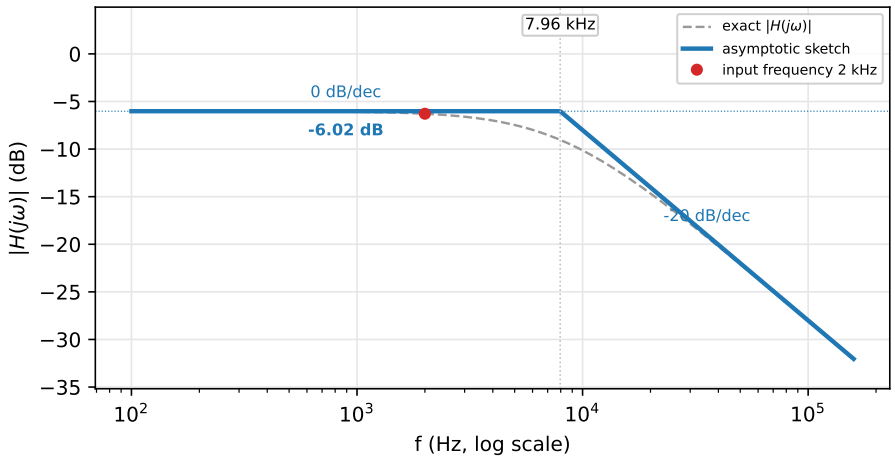
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	50000	7.96 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.5 = -6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 7.96$ kHz	0 dB/dec (flat)	-6.02 dB at end
$f > 7.96$ kHz	−20 dB/dec	-6.02 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.5 = -6.02\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.5	0.5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 7.96\text{k}}}$	0.9698	−0.246
total	**0.4849** (product)	**−0.246** (sum)

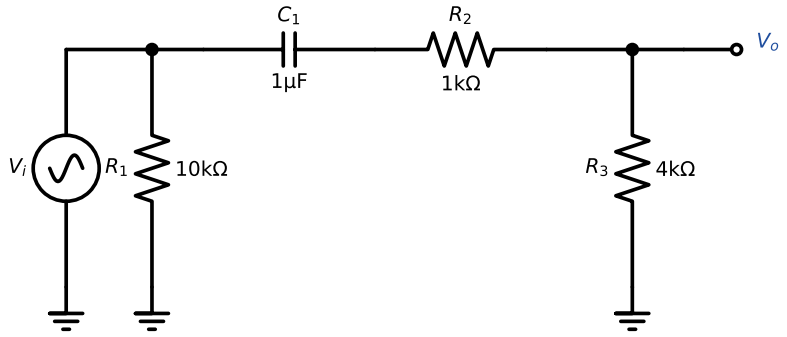
$$|H(j\omega_{in})| = 0.485, \qquad \angle H(j\omega_{in}) = -0.246 \text{ rad}$$
$$v_{o,ac}(t) = 0.485 \times 1 \sin(2\pi \cdot 2000\,t - 0.246) = 0.485 \sin(2\pi \cdot 2000\,t - 0.246)$$

$$\boxed{V_o(t) = 0.485 \sin(2\pi \cdot 2000\,t - 0.246) \text{ V}}$$

Answer

$$H(s) = 0.5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 7.96\text{k}}} \quad \text{(corners: 7.96 kHz)}$$
$$V_o(t) = 0.485 \sin(2\pi \cdot 2000\,t - 0.246) \text{ V}$$

Solution 1A.6 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

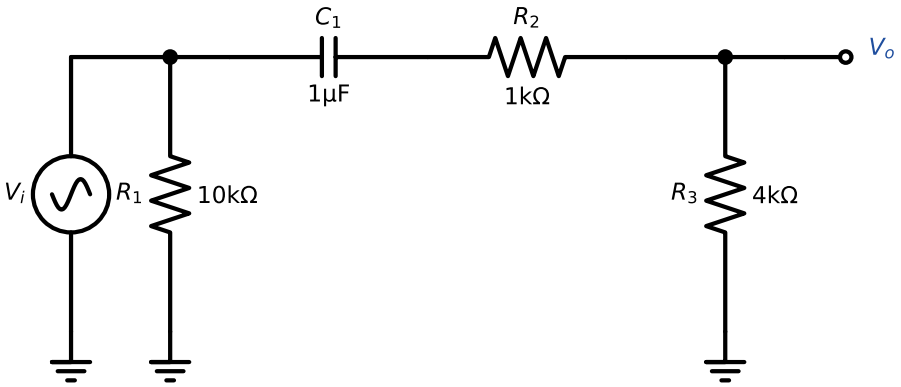
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1 sits directly across the ideal input source V_i. An ideal voltage source keeps its voltage whatever you connect to it, so R_1 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_3}{\frac{1}{sC_1} + R_2 + R_3} = \frac{0.8\,s}{s + 200}$$

So:

$$H(s) = \frac{0.8\,s}{s + 200}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 200$; $s + 200 = 0 \Rightarrow s = -200$.

Factored form (leading coefficient in front):

$$H(s) = 0.8\,\frac{s}{(s + 200)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.8\,\frac{s}{s + 2\pi \cdot 31.8}, \qquad K = 0.8$$

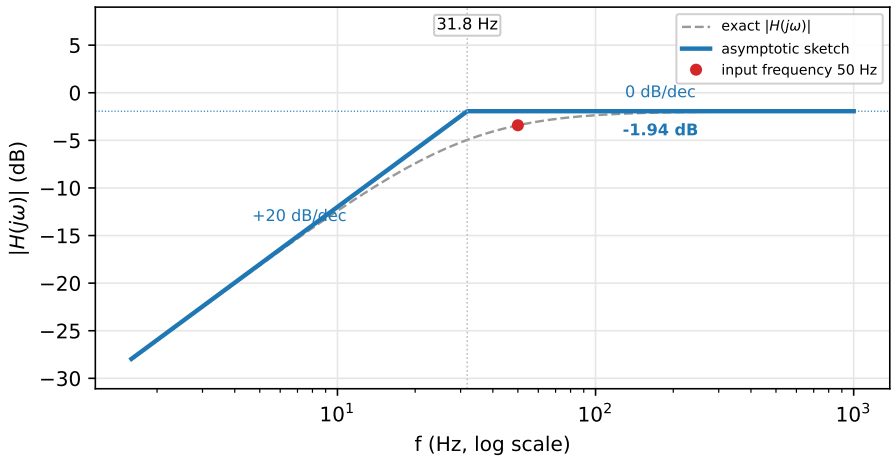
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	200	31.8 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.004\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 31.8$ Hz	+20 dB/dec	-1.94 dB at end
$f > 31.8$ Hz	0 dB/dec (flat)	-1.94 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.8 = -1.94$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 50\,t)$, so $\omega_{in} = 2\pi \cdot 50 = 314\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.8	0.8	0
$\dfrac{s}{s+2\pi \cdot 31.8}$	0.8436	0.567
total	**0.6749** (product)	**0.567** (sum)

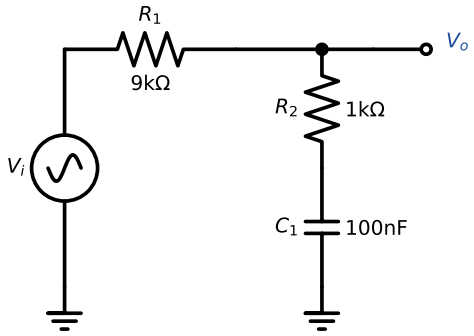
$$|H(j\omega_{in})| = 0.675, \qquad \angle H(j\omega_{in}) = 0.567 \text{ rad}$$
$$v_{o,ac}(t) = 0.675 \times 0.2 \sin(2\pi \cdot 50\,t + 0.567) = 0.135 \sin(2\pi \cdot 50\,t + 0.567)$$

$$\boxed{V_o(t) = 0.135 \sin(2\pi \cdot 50\,t + 0.567)\ \text{V}}$$

Answer

$$H(s) = 0.8\,\frac{s}{s+2\pi \cdot 31.8} \quad \text{(corners: 31.8 Hz)}$$
$$V_o(t) = 0.135 \sin(2\pi \cdot 50\,t + 0.567)\ \text{V}$$

Solution 1A.7 **Medium**



Step 1 — Identify the sources

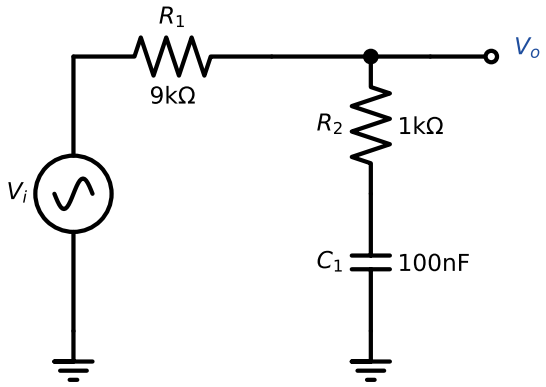
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{0.1\,s + 1000}{s + 1000}$$

So:

$$H(s) = \frac{0.1\,s + 1000}{s + 1000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.1\,s + 1000$; divide by the leading coefficient 0.1 (this does not move the roots): $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.
- Denominator: $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.

Factored form (leading coefficient in front):

$$H(s) = 0.1\,\frac{(s + 10000)}{(s + 1000)}$$

Bode-ready form: rewrite every $(s + p) = p\left(1 + \frac{s}{p}\right)$ and $(s + z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

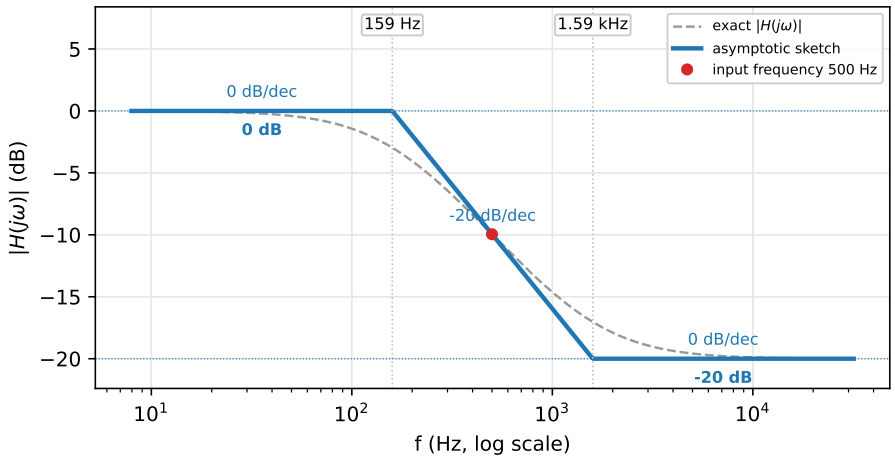
$$H(s) = 1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\,\left(1 + \frac{s}{2\pi \cdot 1.59\text{k}}\right), \qquad K = 1$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
zero	10000	1.59 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 159 Hz	0 dB/dec (flat)	0 dB at end
159 Hz – 1.59 kHz	−20 dB/dec	0 dB at start, -20 dB at end
$f >$ 1.59 kHz	0 dB/dec (flat)	-20 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 500\,t)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 159}}$	0.3033	-1.26
$\left(1+\dfrac{s}{2\pi\cdot 1.59\text{k}}\right)$	1.048	0.304
total	**0.3179** (product)	$-$**0.958** (sum)

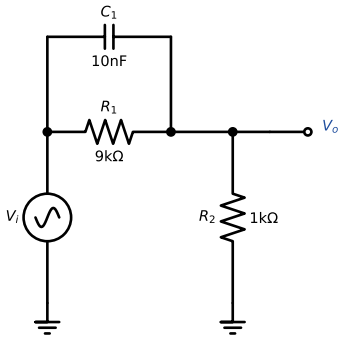
$$|H(j\omega_{in})| = 0.318, \qquad \angle H(j\omega_{in}) = -0.958 \text{ rad}$$
$$v_{o,ac}(t) = 0.318\times 1\sin(2\pi\cdot 500\,t - 0.958) = 0.318\sin(2\pi\cdot 500\,t - 0.958)$$

$$\boxed{V_o(t) = 0.318\sin(2\pi\cdot 500\,t - 0.958)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 159}}\left(1+\frac{s}{2\pi\cdot 1.59\text{k}}\right) \quad \text{(corners: 159 Hz, 1.59 kHz)}$$
$$V_o(t) = 0.318\sin(2\pi\cdot 500\,t - 0.958)\ \text{V}$$

Solution 1A.8 **Medium**



Step 1 — Identify the sources

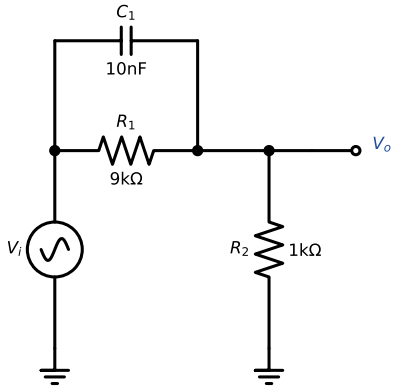
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{R_2}{R_1 \parallel \frac{1}{sC_1} + R_2} = \frac{s + 11100}{s + 1.11 \times 10^5}$$

So:

$$H(s) = \frac{s + 11100}{s + 1.11 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $s + 11100$; $s + 11100 = 0 \Rightarrow s = -11100$.
- Denominator: $s + 1.11 \times 10^5$; $s + 1.11 \times 10^5 = 0 \Rightarrow s = -1.11 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{(s + 11100)}{(s + 1.11 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

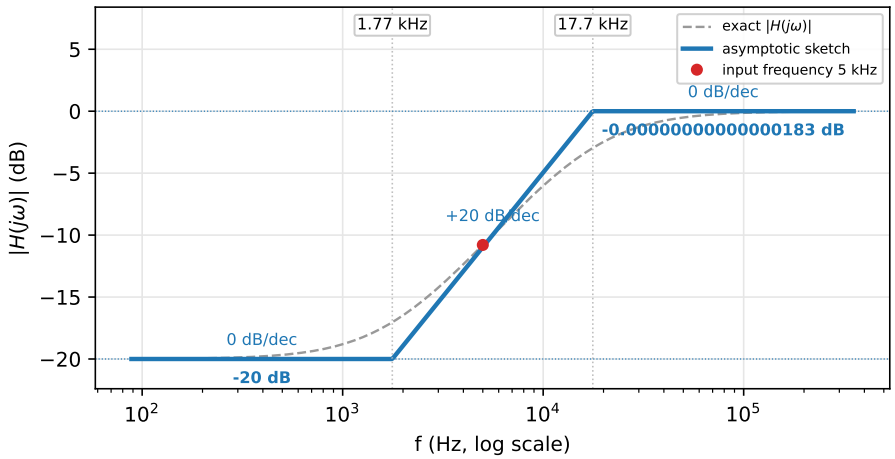
$$H(s) = 0.1\,\left(1 + \frac{s}{2\pi \cdot 1.77\text{k}}\right)\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 17.7\text{k}}}, \qquad K = 0.1$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	11100	1.77 kHz	+20 dB/dec
pole	1.11×10^5	17.7 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.1 = -20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.77$ kHz	0 dB/dec (flat)	-20 dB at end
1.77 kHz – 17.7 kHz	+20 dB/dec	-20 dB at start, 0 dB at end
$f > 17.7$ kHz	0 dB/dec (flat)	0 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.1	0.1	0
$\left(1+\dfrac{s}{2\pi \cdot 1.77\text{k}}\right)$	2.999	1.23
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}}$	0.9623	-0.276
total	**0.2886** (product)	**0.955** (sum)

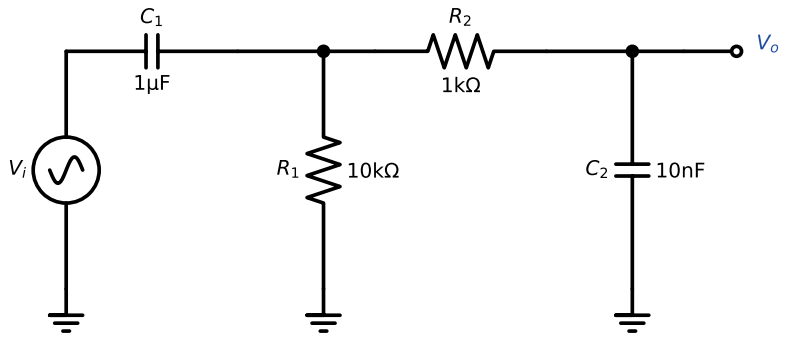
$$|H(j\omega_{in})| = 0.289, \qquad \angle H(j\omega_{in}) = 0.955 \text{ rad}$$
$$v_{o,ac}(t) = 0.289 \times 0.5\,\sin(2\pi \cdot 5000\,t + 0.955) = 0.144\,\sin(2\pi \cdot 5000\,t + 0.955)$$

$$\boxed{V_o(t) = 0.144\,\sin(2\pi \cdot 5000\,t + 0.955)\ \text{V}}$$

Answer

$$H(s) = 0.1\left(1+\frac{s}{2\pi \cdot 1.77\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 17.7\text{k}}} \quad \text{(corners: 1.77 kHz, 17.7 kHz)}$$
$$V_o(t) = 0.144\,\sin(2\pi \cdot 5000\,t + 0.955)\ \text{V}$$

Solution 1A.9 **Medium**



Step 1 — Identify the sources

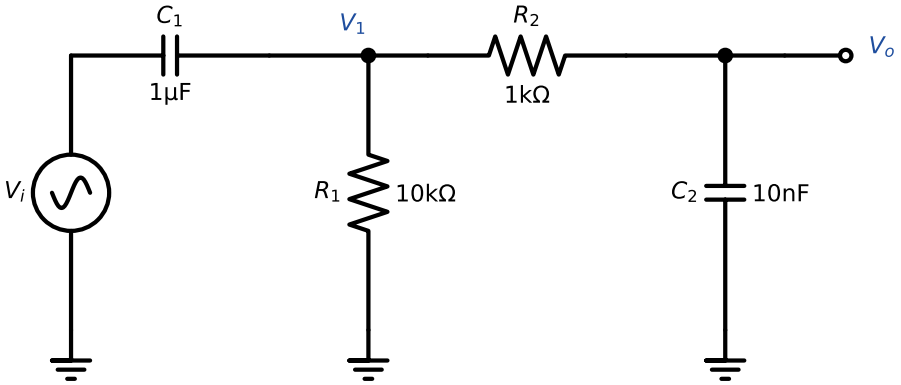
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{909\,s + 9.09 \times 10^7}{s + 9090}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10^5\, s}{s^2 + 1.01 \times 10^5\, s + 10^7}$$
$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{10^5}{s + 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^5\, s}{s^2 + 1.01 \times 10^5\, s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1.01 \times 10^5\, s + 10^7$; quadratic formula:

$$s = \frac{-1.01 \times 10^5 \pm \sqrt{(1.01 \times 10^5)^2 - 4(10^7)}}{2} = \frac{-1.01 \times 10^5 \pm 1.01 \times 10^5}{2} \Rightarrow s = -99,\ -1.01 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^5\, \frac{s}{(s + 99)(s + 1.01 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.99\, \frac{s}{s + 2\pi \cdot 15.8}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 16.1\text{k}}}, \qquad K = 0.99$$

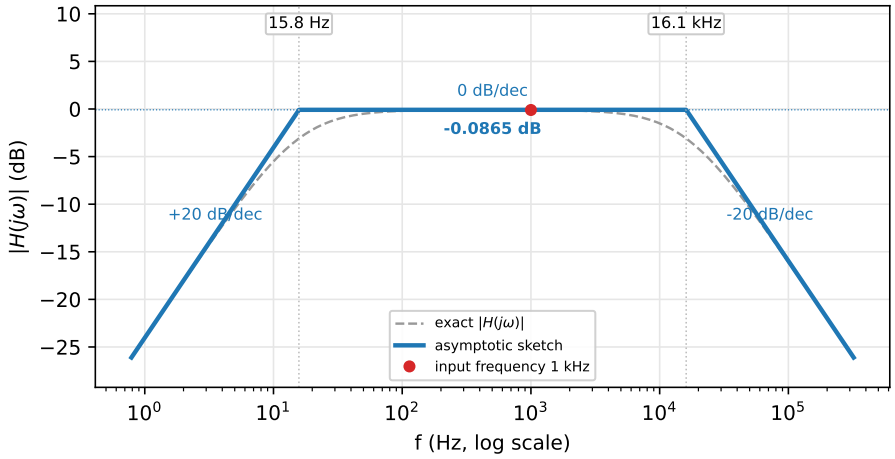
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	99	15.8 Hz	−20 dB/dec
pole	1.01×10^5	16.1 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.01\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 15.8 Hz	+20 dB/dec	-0.0865 dB at end
15.8 Hz – 16.1 kHz	0 dB/dec (flat)	-0.0865 dB at start, -0.0865 dB at end
$f >$ 16.1 kHz	−20 dB/dec	-0.0865 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.99 = -0.0865\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.99	0.9901	0
$\dfrac{s}{s+2\pi\cdot 15.8}$	0.9999	0.0158
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}}$	0.9981	-0.0621
total	**0.9881** (product)	**−0.0464** (sum)

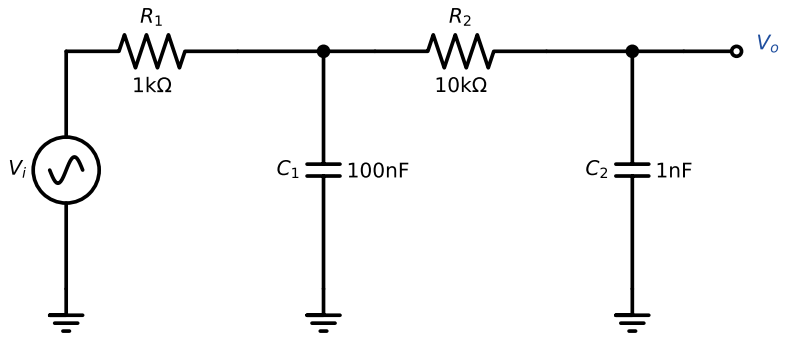
$$|H(j\omega_{in})| = 0.988, \qquad \angle H(j\omega_{in}) = -0.0464 \text{ rad}$$
$$v_{o,ac}(t) = 0.988\times 1\,\sin(2\pi\cdot 1000\,t - 0.0464) = 0.988\,\sin(2\pi\cdot 1000\,t - 0.0464)$$

$$\boxed{V_o(t) = 0.988\,\sin(2\pi\cdot 1000\,t - 0.0464)\ \text{V}}$$

Answer

$$H(s) = 0.99\,\frac{s}{s+2\pi\cdot 15.8}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}} \quad \text{(corners: 15.8 Hz, 16.1 kHz)}$$
$$V_o(t) = 0.988\,\sin(2\pi\cdot 1000\,t - 0.0464)\ \text{V}$$

Solution 1A.10 **Medium**



Step 1 — Identify the sources

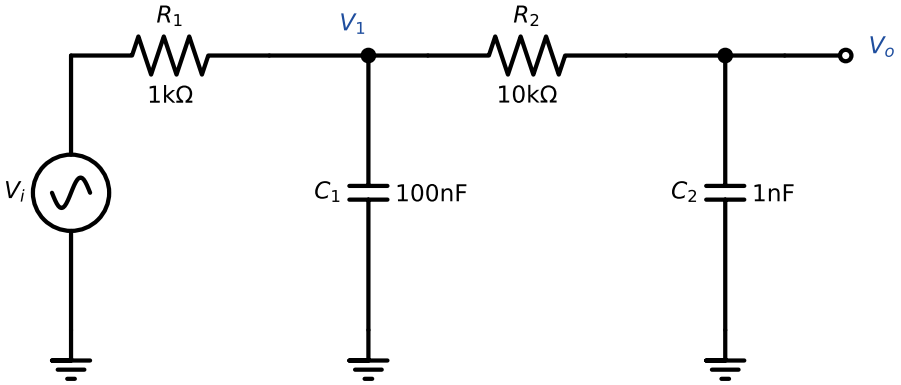
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{10^7\, s + 10^{12}}{s^2 + 1.01 \times 10^5\, s}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{R_1 + Z_1} = \frac{10000\,s + 10^9}{s^2 + 1.11 \times 10^5\,s + 10^9}$$
$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{10^5}{s + 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^9}{s^2 + 1.11 \times 10^5\,s + 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 1.11 \times 10^5\,s + 10^9$; quadratic formula:

$$s = \frac{-1.11 \times 10^5 \pm \sqrt{(1.11 \times 10^5)^2 - 4(10^9)}}{2} = \frac{-1.11 \times 10^5 \pm 91200}{2} \Rightarrow s = -9890,\ -1.01 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^9\,\frac{1}{(s + 9890)(s + 1.01 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{1}{1 + \frac{s}{2\pi \cdot 1.57\text{k}}}\,\frac{1}{1 + \frac{s}{2\pi \cdot 16.1\text{k}}}, \qquad K = 1$$

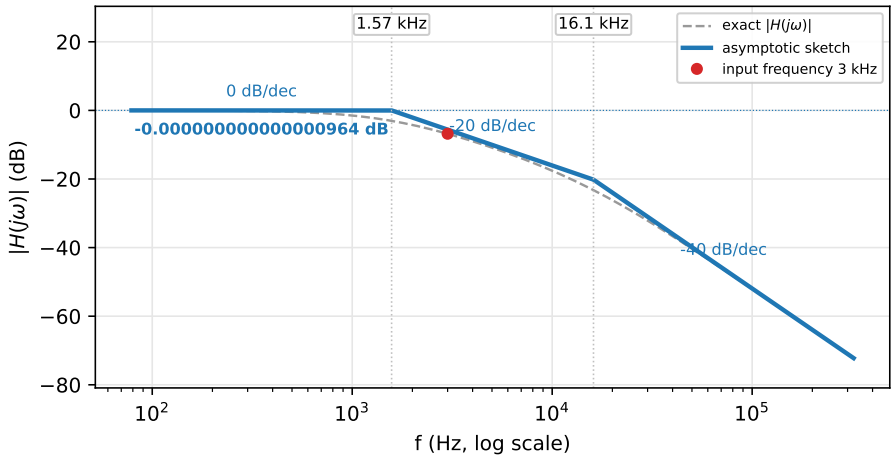
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	9890	1.57 kHz	−20 dB/dec
pole	1.01×10^5	16.1 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.00000000000000964$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.57$ kHz	0 dB/dec (flat)	0 dB at end
1.57 kHz – 16.1 kHz	−20 dB/dec	0 dB at start, -20.2 dB at end
$f > 16.1$ kHz	−40 dB/dec	-20.2 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}1 = -0.000000000000000964\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 3000\,t)$, so $\omega_{in} = 2\pi\cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.57\text{k}}}$	0.4646	-1.09
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}}$	0.9831	-0.184
total	**0.4568** (product)	**−1.27** (sum)

$$|H(j\omega_{in})| = 0.457, \qquad \angle H(j\omega_{in}) = -1.27\ \text{rad}$$
$$v_{o,ac}(t) = 0.457\times 1\,\sin(2\pi\cdot 3000\,t - 1.27) = 0.457\,\sin(2\pi\cdot 3000\,t - 1.27)$$

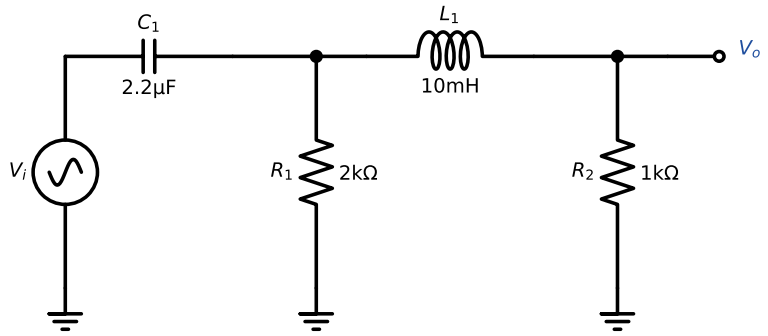
$$\boxed{V_o(t) = 0.457\,\sin(2\pi\cdot 3000\,t - 1.27)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.57\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}} \quad \text{(corners: 1.57 kHz, 16.1 kHz)}$$

$$V_o(t) = 0.457 \sin\left(2\pi \cdot 3000\, t - 1.27\right) \text{ V}$$

Solution 1A.11 **Medium**



Step 1 — Identify the sources

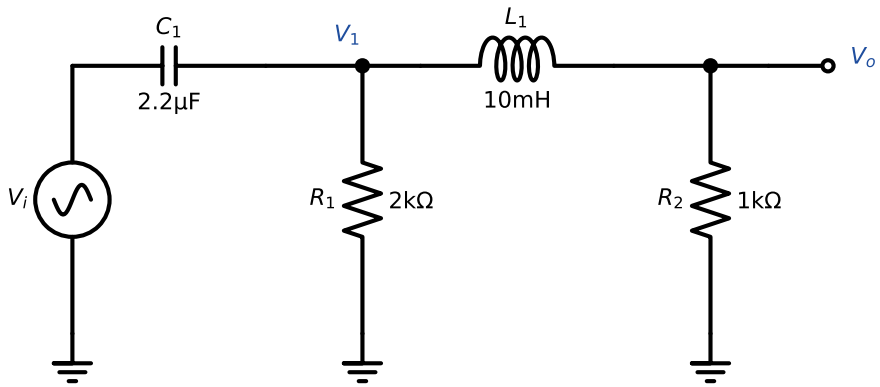
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel (sL_1 + R_2) = \frac{2000\,s + 2 \times 10^8}{s + 3 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10^5\, s}{s^2 + 10^5\, s + 6.82 \times 10^7}$$
$$\frac{V_o}{V_1} = \frac{R_2}{sL_1 + R_2} = \frac{10^5}{s + 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^5\, s}{s^2 + 10^5\, s + 6.82 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 10^5\, s + 6.82 \times 10^7$; quadratic formula:

$$s = \frac{-10^5 \pm \sqrt{(10^5)^2 - 4(6.82 \times 10^7)}}{2} = \frac{-10^5 \pm 98900}{2} \Rightarrow s = -685,\ -99500$$

Factored form (leading coefficient in front):

$$H(s) = 10^5\, \frac{s}{(s + 685)(s + 99500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\, \frac{s}{s + 2\pi \cdot 109}\, \frac{1}{1 + \frac{s}{2\pi \cdot 15.8\text{k}}}, \qquad K = 1$$

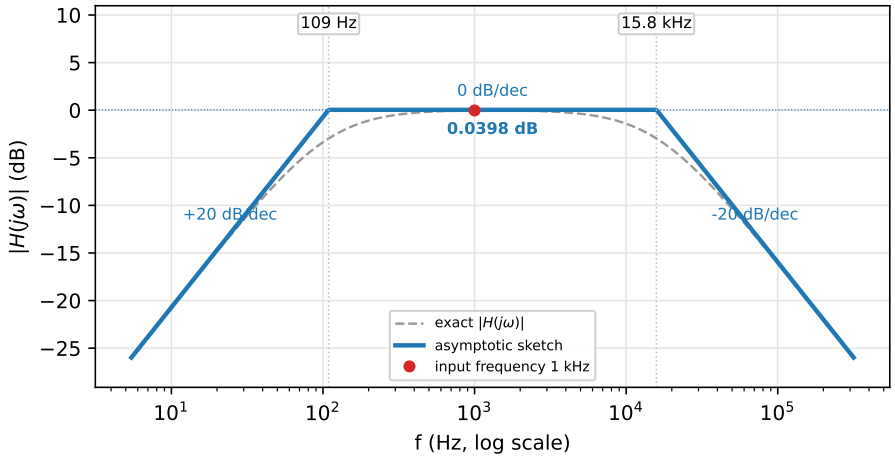
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	685	109 Hz	−20 dB/dec
pole	99500	15.8 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00147\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 109$ Hz	+20 dB/dec	0.0398 dB at end
109 Hz – 15.8 kHz	0 dB/dec (flat)	0.0398 dB at start, 0.0398 dB at end
$f > 15.8$ kHz	−20 dB/dec	0.0398 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}1 = 0.0398\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 2\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1.005	0
$\dfrac{s}{s+2\pi\cdot 109}$	0.9941	0.109
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.8\text{k}}}$	0.998	−0.063
total	**0.9967** (product)	**0.0455** (sum)

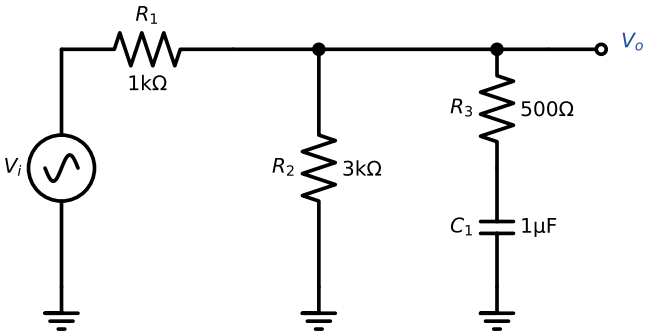
$$|H(j\omega_{in})| = 0.997, \qquad \angle H(j\omega_{in}) = 0.0455 \text{ rad}$$
$$v_{o,ac}(t) = 0.997 \times 2\,\sin(2\pi\cdot 1000\,t + 0.0455) = 1.99\,\sin(2\pi\cdot 1000\,t + 0.0455)$$

$$\boxed{V_o(t) = 1.99\,\sin(2\pi\cdot 1000\,t + 0.0455)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 109}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 15.8\text{k}}} \quad \text{(corners: 109 Hz, 15.8 kHz)}$$
$$V_o(t) = 1.99\,\sin(2\pi\cdot 1000\,t + 0.0455)\ \text{V}$$

Solution 1A.12 **Medium**



Step 1 — Identify the sources

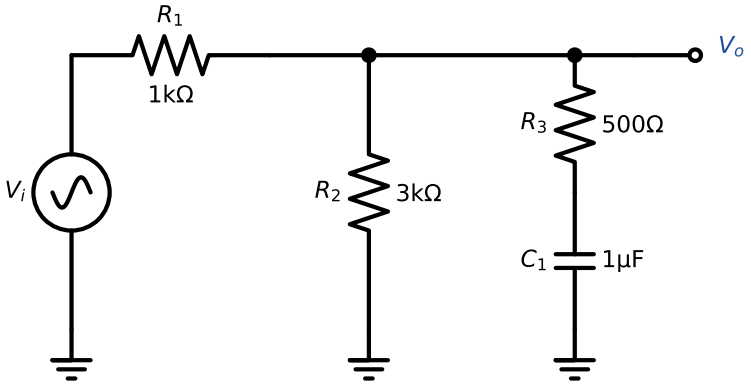
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \left(R_3 + \frac{1}{sC_1}\right) = \frac{429\,s + 8.57 \times 10^5}{s + 286}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{0.3\,s + 600}{s + 800}$$

So:

$$H(s) = \frac{0.3\,s + 600}{s + 800}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.3\,s + 600$; divide by the leading coefficient 0.3 (this does not move the roots): $s + 2000$; $s + 2000 = 0 \Rightarrow s = -2000$.
- Denominator: $s + 800$; $s + 800 = 0 \Rightarrow s = -800$.

Factored form (leading coefficient in front):

$$H(s) = 0.3\,\frac{(s + 2000)}{(s + 800)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

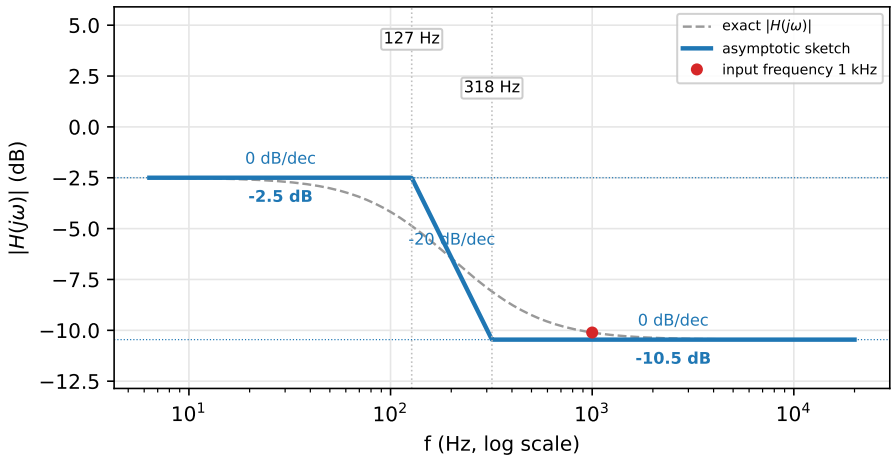
$$H(s) = 0.75\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 127}}\,\left(1 + \frac{s}{2\pi \cdot 318}\right), \qquad K = 0.75$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	800	127 Hz	−20 dB/dec
zero	2000	318 Hz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.75 = -2.5\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 127$ Hz	0 dB/dec (flat)	-2.5 dB at end
127 Hz – 318 Hz	−20 dB/dec	-2.5 dB at start, -10.5 dB at end
$f > 318$ Hz	0 dB/dec (flat)	-10.5 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.75	0.75	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 127}}$	0.1263	-1.44
$\left(1+\dfrac{s}{2\pi \cdot 318}\right)$	3.297	1.26
total	**0.3123** (product)	**−0.182** (sum)

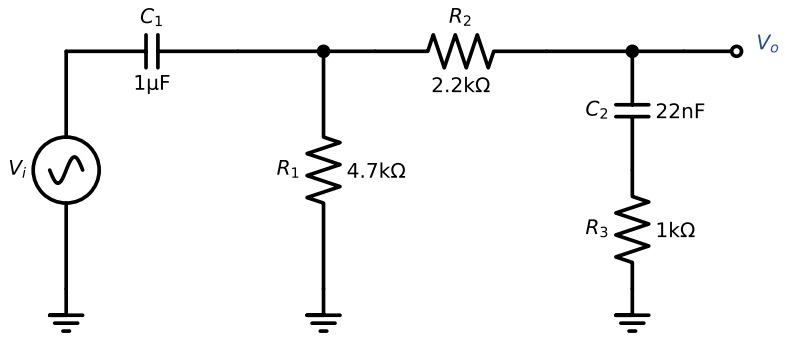
$$|H(j\omega_{in})| = 0.312, \qquad \angle H(j\omega_{in}) = -0.182 \text{ rad}$$
$$v_{o,ac}(t) = 0.312 \times 1 \sin(2\pi \cdot 1000\,t - 0.182) = 0.312 \sin(2\pi \cdot 1000\,t - 0.182)$$

$$\boxed{V_o(t) = 0.312 \sin(2\pi \cdot 1000\,t - 0.182)\ \text{V}}$$

Answer

$$H(s) = 0.75\,\frac{1}{1+\dfrac{s}{2\pi \cdot 127}}\left(1+\frac{s}{2\pi \cdot 318}\right) \quad \text{(corners: 127 Hz, 318 Hz)}$$
$$V_o(t) = 0.312 \sin(2\pi \cdot 1000\,t - 0.182)\ \text{V}$$

Solution 1A.13 **Hard**



Step 1 — Identify the sources

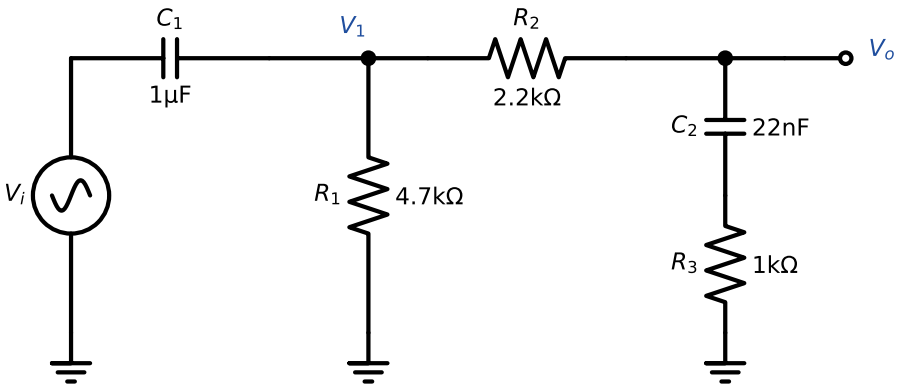
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = \frac{1}{sC_2} + R_3 = \frac{1000\,s + 4.55 \times 10^7}{s}$$

$$Z_1 = R_1 \parallel (R_2 + Z_o) = \frac{1900\,s + 2.7 \times 10^7}{s + 5750}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 14200\,s}{s^2 + 14700\,s + 3.02 \times 10^6}$$
$$\frac{V_o}{V_1} = \frac{Z_o}{R_2 + Z_o} = \frac{0.312\,s + 14200}{s + 14200}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{0.312\,s^2 + 14200\,s}{s^2 + 14700\,s + 3.02 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $0.312\,s + 14200$; divide by the leading coefficient 0.312 (this does not move the roots): $s + 45500$; $s + 45500 = 0 \Rightarrow s = -45500$.
- Denominator: $s^2 + 14700\,s + 3.02 \times 10^6$; quadratic formula:

$$s = \frac{-14700 \pm \sqrt{(14700)^2 - 4(3.02 \times 10^6)}}{2} = \frac{-14700 \pm 14300}{2} \Rightarrow s = -208,\ -14500$$

Factored form (leading coefficient in front):

$$H(s) = 0.312\,\frac{s(s + 45500)}{(s + 208)(s + 14500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

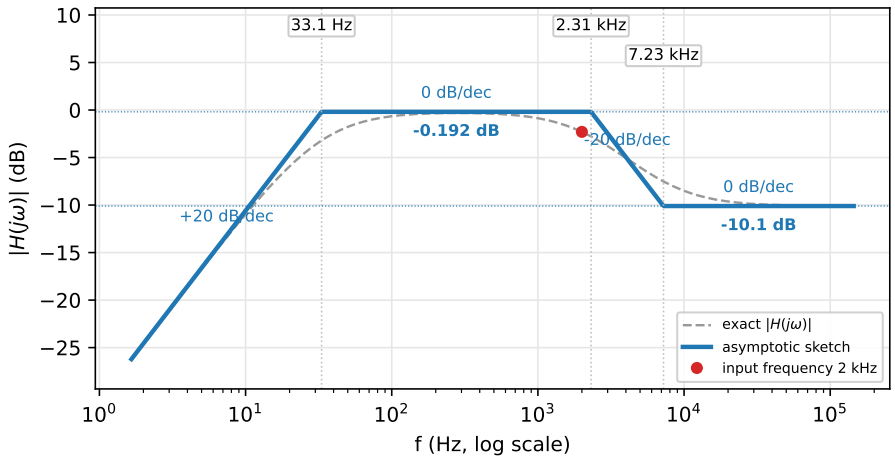
$$H(s) = 0.978\,\frac{s}{s + 2\pi \cdot 33.1}\,\frac{1}{1 + \frac{s}{2\pi \cdot 2.31\text{k}}}\,\left(1 + \frac{s}{2\pi \cdot 7.23\text{k}}\right), \qquad K = 0.978$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	208	33.1 Hz	−20 dB/dec
pole	14500	2.31 kHz	−20 dB/dec
zero	45500	7.23 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0047\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 33.1$ Hz	+20 dB/dec	-0.192 dB at end
33.1 Hz – 2.31 kHz	0 dB/dec (flat)	-0.192 dB at start, -0.192 dB at end
2.31 kHz – 7.23 kHz	−20 dB/dec	-0.192 dB at start, -10.1 dB at end
$f > 7.23$ kHz	0 dB/dec (flat)	-10.1 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\cos\left(2\pi \cdot 2000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.978	0.9782	0
$\dfrac{s}{s + 2\pi \cdot 33.1}$	0.9999	0.0166
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 2.31\text{k}}}$	0.7562	−0.713
$\left(1 + \dfrac{s}{2\pi \cdot 7.23\text{k}}\right)$	1.038	0.27
total	**0.7673** (product)	**−0.427** (sum)

$$|H(j\omega_{in})| = 0.767, \qquad \angle H(j\omega_{in}) = -0.427 \text{ rad}$$
$$v_{o,ac}(t) = 0.767 \times 1 \cos\left(2\pi \cdot 2000\,t + 0.358\right) = 0.767\cos\left(2\pi \cdot 2000\,t + 0.358\right)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-0.427) = 0.358\,\text{rad.}$)

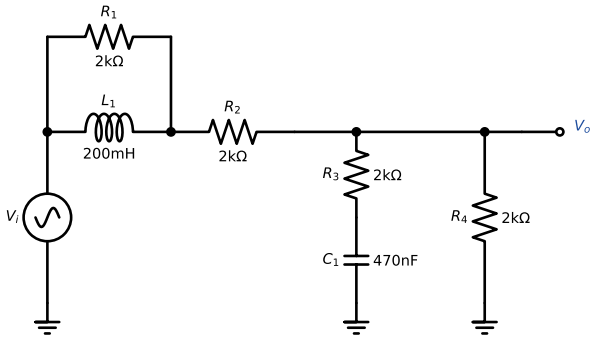
$$\boxed{V_o(t) = 0.767\,\cos\left(2\pi \cdot 2000\,t + 0.358\right)\ \text{V}}$$

Answer

$$H(s) = 0.978\,\frac{s}{s + 2\pi \cdot 33.1}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 2.31\text{k}}}\left(1 + \frac{s}{2\pi \cdot 7.23\text{k}}\right) \quad \text{(corners: 33.1 Hz, 2.31 kHz, 7.23 kHz)}$$

$$V_o(t) = 0.767\,\cos\left(2\pi \cdot 2000\,t + 0.358\right)\ \text{V}$$

Solution 1A.14 **Hard**



Step 1 — Identify the sources

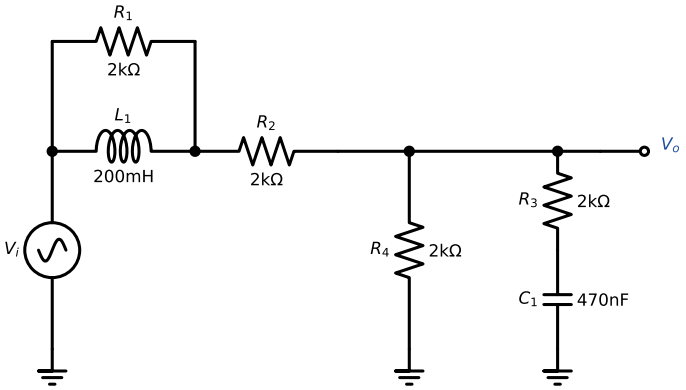
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_4 \parallel \left(R_3 + \frac{1}{sC_1}\right) = \frac{1000\,s + 1.06 \times 10^6}{s + 532}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{(sL_1 \parallel R_1) + R_2 + Z_o} = \frac{0.2\,s^2 + 2210\,s + 2.13 \times 10^6}{s^2 + 6640\,s + 4.26 \times 10^6}$$

So:

$$H(s) = \frac{0.2\,s^2 + 2210\,s + 2.13 \times 10^6}{s^2 + 6640\,s + 4.26 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.2\,s^2 + 2210\,s + 2.13 \times 10^6$; divide by the leading coefficient 0.2 (this does not move the roots): $s^2 + 11100\,s + 1.06 \times 10^7$; quadratic formula:

$$s = \frac{-11100 \pm \sqrt{(11100)^2 - 4(1.06 \times 10^7)}}{2} = \frac{-11100 \pm 8940}{2} \Rightarrow s = -1060,\ -10000$$

- Denominator: $s^2 + 6640\,s + 4.26 \times 10^6$; quadratic formula:

$$s = \frac{-6640 \pm \sqrt{(6640)^2 - 4(4.26 \times 10^6)}}{2} = \frac{-6640 \pm 5200}{2} \Rightarrow s = -719,\ -5920$$

Factored form (leading coefficient in front):

$$H(s) = 0.2\,\frac{(s + 1060)(s + 10000)}{(s + 719)(s + 5920)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

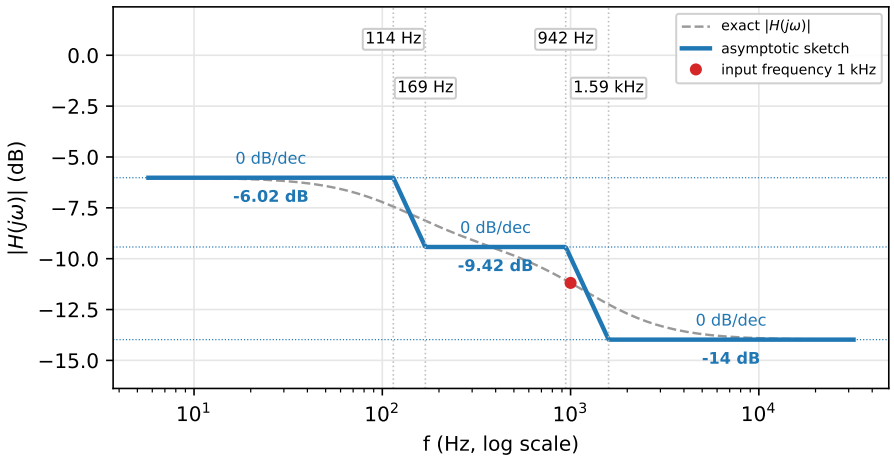
$$H(s) = 0.5\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 114}}\left(1 + \frac{s}{2\pi \cdot 169}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 942}}\left(1 + \frac{s}{2\pi \cdot 1.59\text{k}}\right), \qquad K = 0.5$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	719	114 Hz	−20 dB/dec
zero	1060	169 Hz	+20 dB/dec
pole	5920	942 Hz	−20 dB/dec
zero	10000	1.59 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.5 = -6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 114 Hz	0 dB/dec (flat)	-6.02 dB at end
114 Hz – 169 Hz	−20 dB/dec	-6.02 dB at start, -9.42 dB at end
169 Hz – 942 Hz	0 dB/dec (flat)	-9.42 dB at start, -9.42 dB at end
942 Hz – 1.59 kHz	−20 dB/dec	-9.42 dB at start, -14 dB at end
$f >$ 1.59 kHz	0 dB/dec (flat)	-14 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin\left(2\pi\cdot 1000\,t + \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.5	0.5	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 114}}$	0.1137	-1.46
$\left(1+\dfrac{s}{2\pi\cdot 169}\right)$	5.99	1.4
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 942}}$	0.6857	-0.815
$\left(1+\dfrac{s}{2\pi\cdot 1.59\text{k}}\right)$	1.181	0.561
total	**0.2757** (product)	**−0.308** (sum)

$$|H(j\omega_{in})| = 0.276, \qquad \angle H(j\omega_{in}) = -0.308 \text{ rad}$$
$$v_{o,ac}(t) = 0.276 \times 1 \sin\left(2\pi\cdot 1000\,t + 0.739\right) = 0.276\sin\left(2\pi\cdot 1000\,t + 0.739\right)$$

(phase: $\theta + \angle H = \frac{\pi}{3} + (-0.308) = 0.739\,\text{rad}$.)

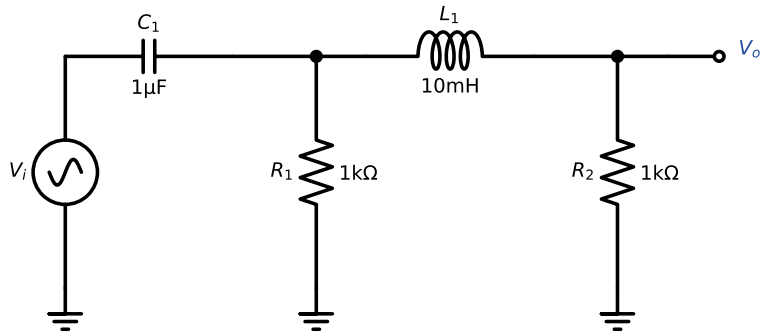
$$\boxed{V_o(t) = 0.276\sin\left(2\pi\cdot 1000\,t + 0.739\right)\ \text{V}}$$

Answer

$H(s) = 0.5\,\dfrac{1}{1+\dfrac{s}{2\pi\cdot 114}}\left(1+\dfrac{s}{2\pi\cdot 169}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 942}}\left(1+\dfrac{s}{2\pi\cdot 1.59\text{k}}\right)$ (corners: 114 Hz, 169 Hz, 942 Hz, 1.59 kHz)

$V_o(t) = 0.276\,\sin\left(2\pi\cdot 1000\,t + 0.739\right)\ \text{V}$

Solution 1A.15 **Hard**



Step 1 — Identify the sources

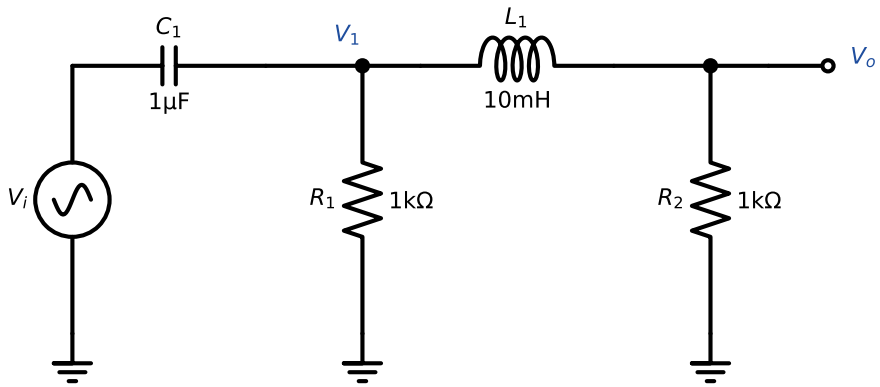
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel (sL_1 + R_2) = \frac{1000\,s + 10^8}{s + 2 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10^5\, s}{s^2 + 1.01 \times 10^5\, s + 2 \times 10^8}$$
$$\frac{V_o}{V_1} = \frac{R_2}{sL_1 + R_2} = \frac{10^5}{s + 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^5\, s}{s^2 + 1.01 \times 10^5\, s + 2 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1.01 \times 10^5\, s + 2 \times 10^8$; quadratic formula:

$$s = \frac{-1.01 \times 10^5 \pm \sqrt{(1.01 \times 10^5)^2 - 4(2 \times 10^8)}}{2} = \frac{-1.01 \times 10^5 \pm 97000}{2} \Rightarrow s = -2020,\ -99000$$

Factored form (leading coefficient in front):

$$H(s) = 10^5\, \frac{s}{(s + 2020)(s + 99000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1.01\, \frac{s}{s + 2\pi \cdot 322}\, \frac{1}{1 + \frac{s}{2\pi \cdot 15.8\text{k}}}, \qquad K = 1.01$$

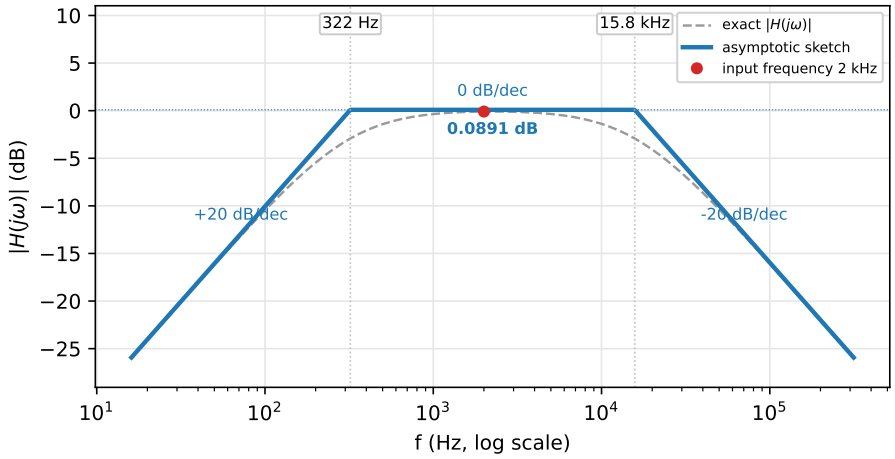
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2020	322 Hz	−20 dB/dec
pole	99000	15.8 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 5 \times 10^{-4}\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 322$ Hz	+20 dB/dec	0.0891 dB at end
322 Hz – 15.8 kHz	0 dB/dec (flat)	0.0891 dB at start, 0.0891 dB at end
$f > 15.8$ kHz	−20 dB/dec	0.0891 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}1.01 = 0.0891\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.01	1.01	0
$\dfrac{s}{s+2\pi\cdot 322}$	0.9873	0.159
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.8\text{k}}}$	0.992	-0.126
total	**0.9896** (product)	**0.0331** (sum)

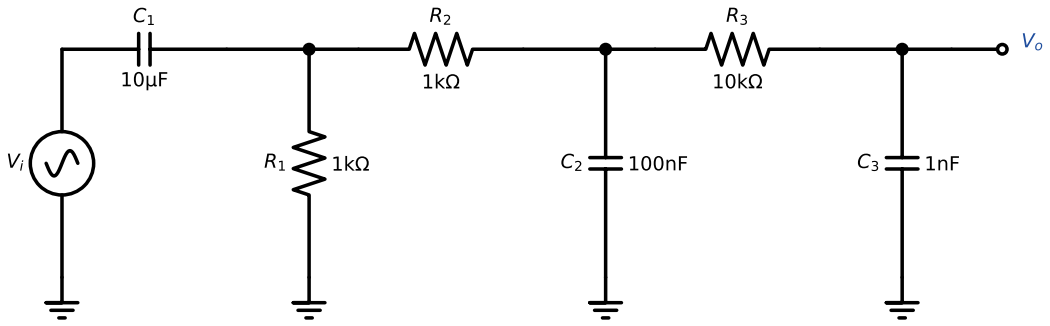
$$|H(j\omega_{in})| = 0.99, \qquad \angle H(j\omega_{in}) = 0.0331\ \text{rad}$$
$$v_{o,ac}(t) = 0.99\times 0.5\,\sin(2\pi\cdot 2000\,t + 0.0331) = 0.495\,\sin(2\pi\cdot 2000\,t + 0.0331)$$

$$\boxed{V_o(t) = 0.495\,\sin(2\pi\cdot 2000\,t + 0.0331)\ \text{V}}$$

Answer

$$H(s) = 1.01\,\frac{s}{s+2\pi\cdot 322}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 15.8\text{k}}} \quad \text{(corners: 322 Hz, 15.8 kHz)}$$
$$V_o(t) = 0.495\,\sin(2\pi\cdot 2000\,t + 0.0331)\ \text{V}$$

Solution 1A.16 **Hard**



Step 1 — Identify the sources

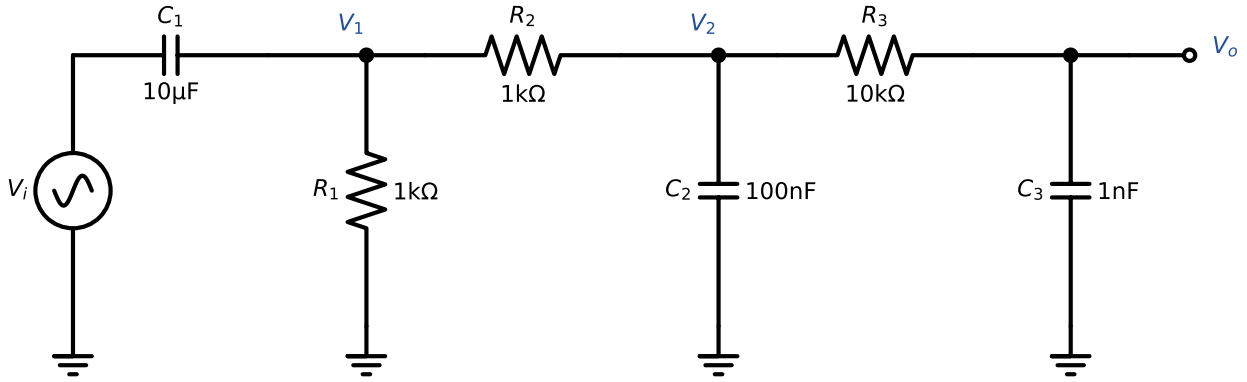
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_2 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_2}{V_1} \cdot \frac{V_o}{V_2}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_2 = \frac{1}{sC_2} \parallel \left(R_3 + \frac{1}{sC_3}\right) = \frac{10^7\, s + 10^{12}}{s^2 + 1.01 \times 10^5\, s}$$

$$Z_1 = R_1 \parallel (R_2 + Z_2) = \frac{500\, s^2 + 5.55 \times 10^7\, s + 5 \times 10^{11}}{s^2 + 1.06 \times 10^5\, s + 5 \times 10^8}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^3 + 1.11 \times 10^5\, s^2 + 10^9\, s}{s^3 + 1.11 \times 10^5\, s^2 + 1.02 \times 10^9\, s + 10^{11}}$$
$$\frac{V_2}{V_1} = \frac{Z_2}{R_2 + Z_2} = \frac{10000\, s + 10^9}{s^2 + 1.11 \times 10^5\, s + 10^9}$$
$$\frac{V_o}{V_2} = \frac{\frac{1}{sC_3}}{R_3 + \frac{1}{sC_3}} = \frac{10^5}{s + 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^9\, s}{s^3 + 1.11 \times 10^5\, s^2 + 1.02 \times 10^9\, s + 10^{11}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^3 + 1.11 \times 10^5\, s^2 + 1.02 \times 10^9\, s + 10^{11}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -99,\ -9990,\ -1.01 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 10^9\, \frac{s}{(s + 99)(s + 9990)(s + 1.01 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.99\, \frac{s}{s + 2\pi \cdot 15.8}\, \frac{1}{1 + \frac{s}{2\pi \cdot 1.59\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 16.1\text{k}}}, \qquad K = 0.99$$

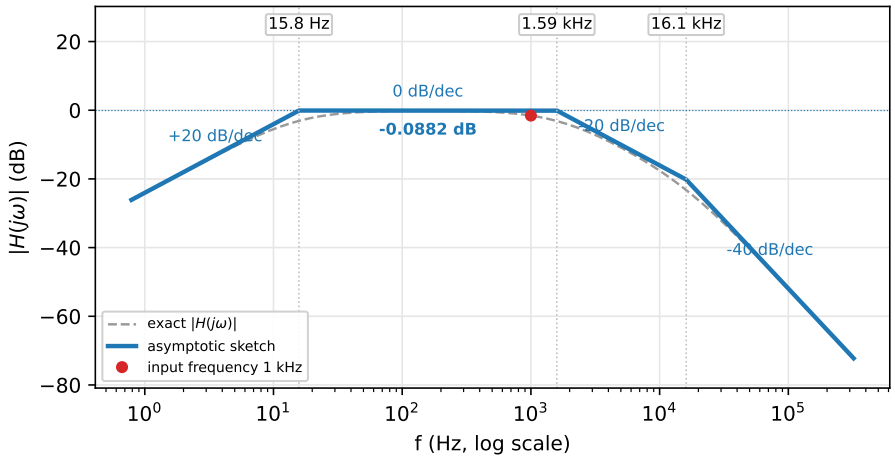
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	99	15.8 Hz	−20 dB/dec
pole	9990	1.59 kHz	−20 dB/dec
pole	1.01×10^5	16.1 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.01\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.8$ Hz	+20 dB/dec	-0.0882 dB at end
15.8 Hz – 1.59 kHz	0 dB/dec (flat)	-0.0882 dB at start, -0.0882 dB at end
1.59 kHz – 16.1 kHz	−20 dB/dec	-0.0882 dB at start, -20.2 dB at end
$f > 16.1$ kHz	−40 dB/dec	-20.2 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.99 = -0.0882\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.99	0.9899	0
$\dfrac{s}{s+2\pi\cdot 15.8}$	0.9999	0.0158
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.8465	−0.561
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}}$	0.9981	−0.0621
total	**0.8363** (product)	**−0.608** (sum)

$$|H(j\omega_{in})| = 0.836, \qquad \angle H(j\omega_{in}) = -0.608 \text{ rad}$$

$$v_{o,ac}(t) = 0.836 \times 1 \sin\left(2\pi \cdot 1000\, t - 0.608\right) = 0.836 \sin\left(2\pi \cdot 1000\, t - 0.608\right)$$

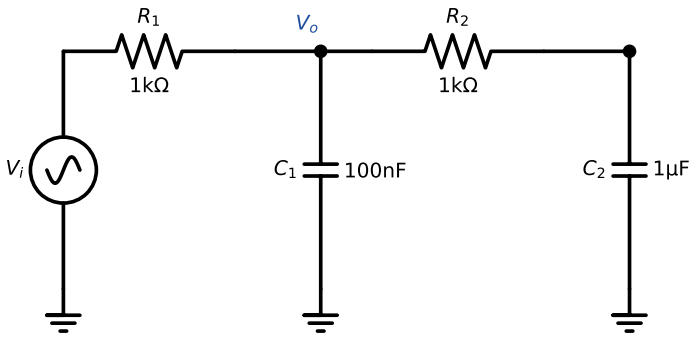
$$\boxed{V_o(t) = 0.836 \sin\left(2\pi \cdot 1000\, t - 0.608\right) \text{ V}}$$

Answer

$$H(s) = 0.99 \frac{s}{s + 2\pi \cdot 15.8} \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 16.1\text{k}}} \quad \text{(corners: 15.8 Hz, 1.59 kHz, 16.1 kHz)}$$

$$V_o(t) = 0.836 \sin\left(2\pi \cdot 1000\, t - 0.608\right) \text{ V}$$

Solution 1A.17 **Hard**



Step 1 — Identify the sources

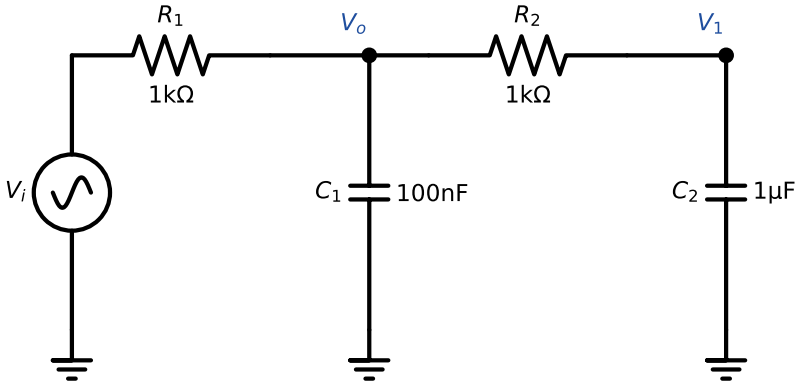
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{10^7\, s + 10^{10}}{s^2 + 11000\, s}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 + Z_o} = \frac{10000\, s + 10^7}{s^2 + 21000\, s + 10^7}$$

So:

$$H(s) = \frac{10000\,s + 10^7}{s^2 + 21000\,s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $10000\,s + 10^7$; divide by the leading coefficient 10000 (this does not move the roots): $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.
- Denominator: $s^2 + 21000\,s + 10^7$; quadratic formula:

$$s = \frac{-21000 \pm \sqrt{(21000)^2 - 4(10^7)}}{2} = \frac{-21000 \pm 20000}{2} \Rightarrow s = -488,\ -20500$$

Factored form (leading coefficient in front):

$$H(s) = 10000\,\frac{(s + 1000)}{(s + 488)(s + 20500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

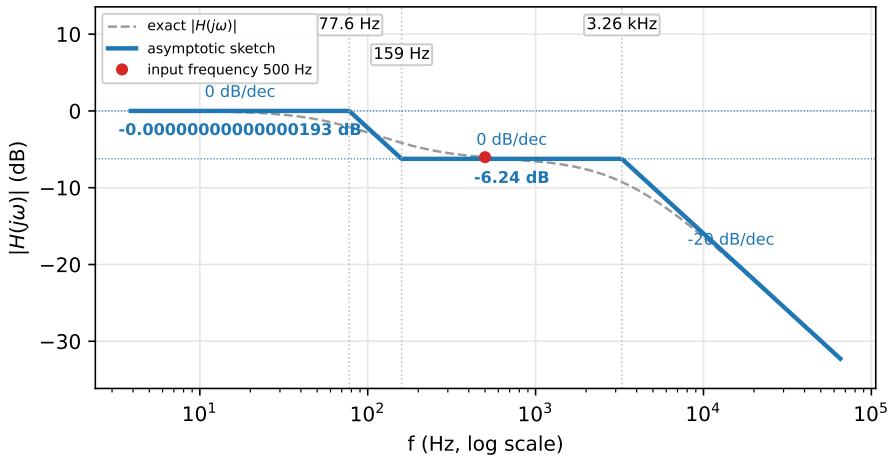
$$H(s) = 1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 77.6}}\,\left(1 + \frac{s}{2\pi \cdot 159}\right)\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.26\text{k}}}, \qquad K = 1$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	488	77.6 Hz	−20 dB/dec
zero	1000	159 Hz	+20 dB/dec
pole	20500	3.26 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.00000000000000193\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 77.6 Hz	0 dB/dec (flat)	0 dB at end
77.6 Hz – 159 Hz	−20 dB/dec	0 dB at start, -6.24 dB at end
159 Hz – 3.26 kHz	0 dB/dec (flat)	-6.24 dB at start, -6.24 dB at end
$f >$ 3.26 kHz	−20 dB/dec	-6.24 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 77.6}}$	0.1533	-1.42
$\left(1+\dfrac{s}{2\pi \cdot 159}\right)$	3.297	1.26
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 3.26\text{k}}}$	0.9885	-0.152
total	**0.4997** (product)	**−0.306** (sum)

$$|H(j\omega_{in})| = 0.5, \qquad \angle H(j\omega_{in}) = -0.306 \text{ rad}$$
$$v_{o,ac}(t) = 0.5 \times 1 \sin(2\pi \cdot 500\,t - 0.306) = 0.5\sin(2\pi \cdot 500\,t - 0.306)$$

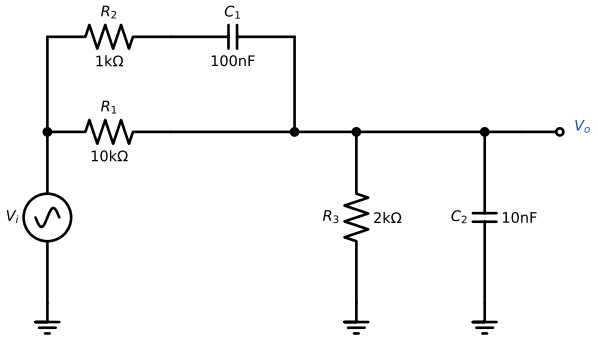
$$\boxed{V_o(t) = 0.5\sin(2\pi \cdot 500\,t - 0.306)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 77.6}}\left(1+\frac{s}{2\pi \cdot 159}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 3.26\text{k}}} \quad \text{(corners: 77.6 Hz, 159 Hz, 3.26 kHz)}$$

$$V_o(t) = 0.5 \sin\left(2\pi \cdot 500\, t - 0.306\right) \text{ V}$$

Solution 1A.18 **Hard**



Step 1 — Identify the sources

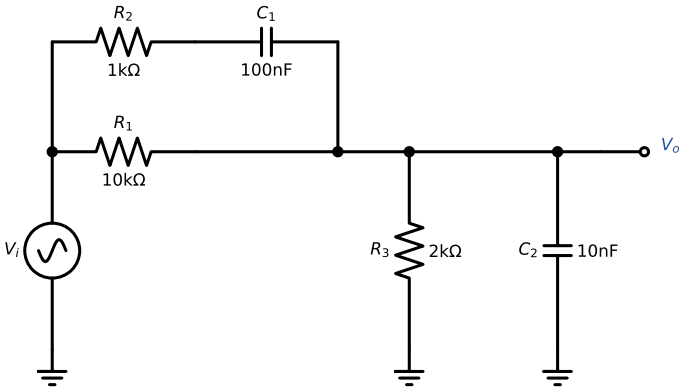
Independent sources: the AC input $V_i(t)$. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_o$. There is a single link, so $H(s)$ is just one voltage divider (tool T2). Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_3 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 50000}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_o}{V_i} = \frac{Z_o}{R_1 \parallel \left(R_2 + \frac{1}{sC_1}\right) + Z_o} = \frac{1.1 \times 10^5\, s + 10^8}{s^2 + 1.7 \times 10^5\, s + 6 \times 10^8}$$

So:

$$H(s) = \frac{1.1 \times 10^5\, s + 10^8}{s^2 + 1.7 \times 10^5\, s + 6 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $1.1 \times 10^5\, s + 10^8$; divide by the leading coefficient 1.1×10^5 (this does not move the roots): $s + 909$; $s + 909 = 0 \Rightarrow s = -909$.
- Denominator: $s^2 + 1.7 \times 10^5\, s + 6 \times 10^8$; quadratic formula:

$$s = \frac{-1.7 \times 10^5 \pm \sqrt{(1.7 \times 10^5)^2 - 4(6 \times 10^8)}}{2} = \frac{-1.7 \times 10^5 \pm 1.63 \times 10^5}{2} \Rightarrow s = -3610,\ -1.66 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 1.1 \times 10^5\, \frac{(s + 909)}{(s + 3610)(s + 1.66 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

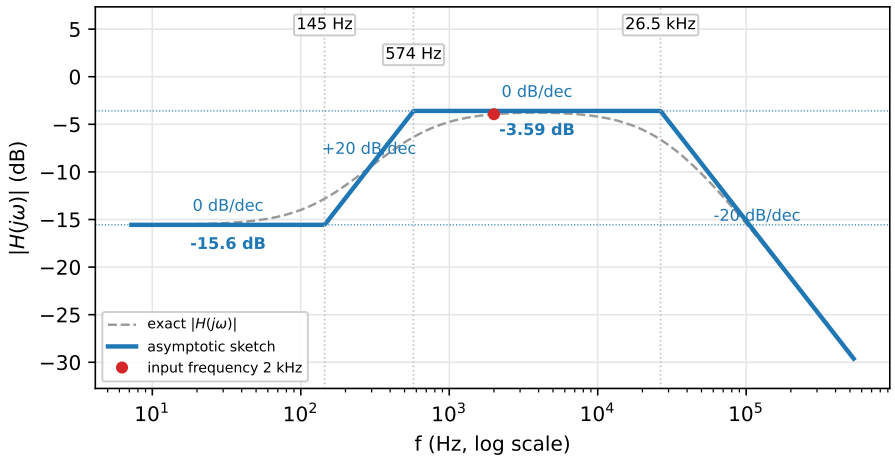
$$H(s) = 0.167\, \left(1 + \frac{s}{2\pi \cdot 145}\right)\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 574}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 26.5\text{k}}}, \qquad K = 0.167$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	909	145 Hz	+20 dB/dec
pole	3610	574 Hz	−20 dB/dec
pole	1.66×10^5	26.5 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.167 = -15.6\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 145 Hz	0 dB/dec (flat)	-15.6 dB at end
145 Hz – 574 Hz	+20 dB/dec	-15.6 dB at start, -3.59 dB at end
574 Hz – 26.5 kHz	0 dB/dec (flat)	-3.59 dB at start, -3.59 dB at end
$f >$ 26.5 kHz	−20 dB/dec	-3.59 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin\left(2\pi\cdot 2000\,t - \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.167	0.1667	0
$\left(1+\dfrac{s}{2\pi\cdot 145}\right)$	13.86	1.5
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 574}}$	0.2758	-1.29
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 26.5\text{k}}}$	0.9972	-0.0754
total	**0.6353** (product)	**0.132** (sum)

$$|H(j\omega_{in})| = 0.635, \qquad \angle H(j\omega_{in}) = 0.132 \text{ rad}$$
$$v_{o,ac}(t) = 0.635 \times 1 \sin\left(2\pi\cdot 2000\,t - 0.392\right) = 0.635\sin\left(2\pi\cdot 2000\,t - 0.392\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{6} + (0.132) = -0.392\,\text{rad}$.)

$$\boxed{V_o(t) = 0.635\sin\left(2\pi\cdot 2000\,t - 0.392\right)\ \text{V}}$$

Answer

$$H(s) = 0.167\left(1 + \frac{s}{2\pi \cdot 145}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 574}}\frac{1}{1 + \dfrac{s}{2\pi \cdot 26.5\text{k}}} \quad \text{(corners: 145 Hz, 574 Hz, 26.5 kHz)}$$

$$V_o(t) = 0.635\,\sin\left(2\pi \cdot 2000\,t - 0.392\right)\ \text{V}$$